APPENDIX C

# Apple II Schematics

This appendix contains schematics for the Apple II mother board, keyboard, and power supply. Mother board schematics are provided for circuits dating from the earliest Apple II to the most recent revision available for publication. The paragraphs that follow contain information intended to help you use the schematics.

## REVISIONS

### Mother Boards

As described in Appendix B, the earliest Apple II mother board is known as Rev. 0. A short time after the Apple's introduction, several circuit changes resulted in the Rev. 1 mother board. Rev. 1 includes a color killer and two more HIRES colors. There were no more significant circuit changes until the introduction of the Rev. 7 mother board. Rev. 7 deletes the Memory Jumper Blocks and changes the character generator ROM. The next significant change involved modification of the mother board and packaging to reduce RFI. Some minor circuit changes were made at the same time. The major revisions of the mother board are denoted in this book as Rev. 0, Rev. 1, Rev. 7, and RFI.

Appendix B contains additional information on the differences between revisions. It also describes how you can examine the mother board to determine the revision of a particular Apple II. Note that the circuit description in the body of this book corresponds to the RFI schematics. Material in Appendix B describes the differences in operation of Revisions 0, 1, and 7 relative to the operation of the RFI mother board.

A notation on each schematic indicates the revision to which the schematic applies. Most schematics apply to two or more revisions. Minor circuit differences between the revisions are indicated by notes or insets on the drawing. When there are major differences between revisions, separate schematics are used for each revision. As you trace a signal from schematic to schematic, be sure to use the drawing that corresponds to the revision of interest.

## Keyboards

The original Apple II keyboard was constructed on a single printed circuit board and is thus referred to as the "single-piece keyboard." Apple later introduced a "two-piece keyboard." The two-piece keyboard contains one printed circuit board on which the keys are mounted. A second printed circuit board containing the ICs then plugs onto the first printed circuit board. This appendix contains schematics for both the single- and two-piece keyboards.

## SYMBOLS

Fig. C-1 shows some of the symbols used on the schematics. When a signal enters or exits a schematic, the signal name is shown beside an arrow symbol (Examples a through d). The direction of the arrow indicates the direction of signal flow. A double-headed arrow indicates bidirectional signal flow, such as on a data bus (Example c). If the signal is part of the system bus, a number inside the arrow symbol gives the peripheral I/O connector pin on which the signal appears (Examples c and d). A "To" or "From" notation beside a signal name indicates the other schematics on which the signal appears. The designations "Part 1" and "Part 2" refer to the two separate drawings used to show some schematics. For example, the video generator schematic is divided into two parts.

Logic functions that are active on a low level are shown by a small circle or

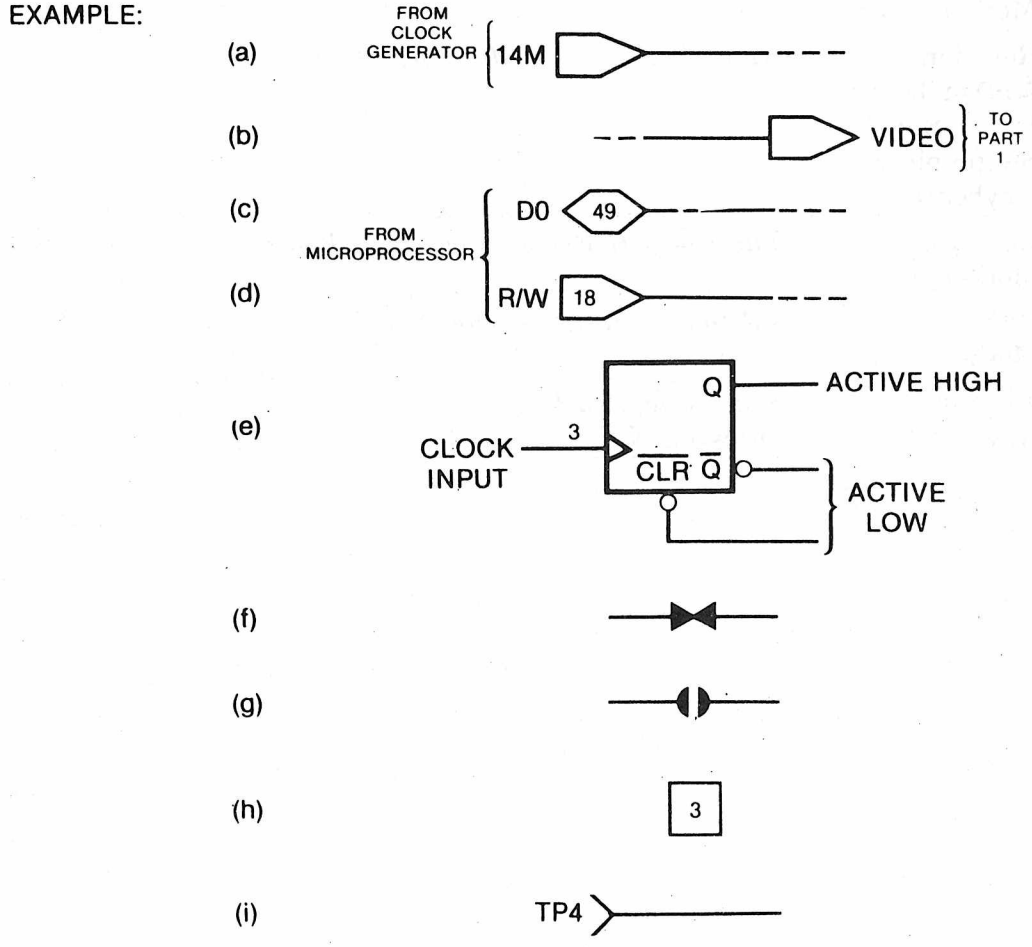



**Fig. C-1.** Schematic symbols.

"bubble." For example, see the $\overline{Q}$ output and $\overline{CLR}$ input in Example e. The symbol at pin 3 is a clock input.

The symbols in Examples f and g represent the shapes of printed circuit board traces that are meant to be altered in order to modify the circuit. The *bow tie* in Example f would be cut between the arrows to open the circuit. The semicircles in Example g would be bridged with solder to close the circuit.

A number in a square (Example h) refers to a note shown elsewhere on the drawing. The symbol in Example i represents a test point or other labeled solder point on the circuit board.

When an IC is powered from +5 volts and ground, and these voltages connect to the usual opposite corner pins of the IC, then the power pins are not shown on the schematic. If the +5 volts connects to an unusual pin or if the IC requires multiple voltages, then the power pins *are* shown. In general, bypass capacitors, ferrite beads, and isolated grounds are not depicted in the drawings.

The Apple II schematics are shown in Figs. C-2* through C-23*.

## ACKNOWLEDGMENTS

The schematics in this appendix have been used with the permission of Apple Computer, Inc. They have been redrawn from the following Apple Computer documents:

| | |
|---|---|
| Revision 0 Mother board | *Apple II Reference Manual* ("Red Book"), Jan. 1978 |
| Revision 1 Mother board, Power Supply, Single-piece Keyboard | *Apple II Reference Manual*, 1979 |
| Revision 7 Mother board | *The Apple II Revision 07 Main Board* |
| RFI Mother board | *Addendum to the Apple II Reference Manual* |
| Two-piece Keyboard | *Schematic, A-2 Keyboard & Encoder*, Drawing No. 050-0021-D |

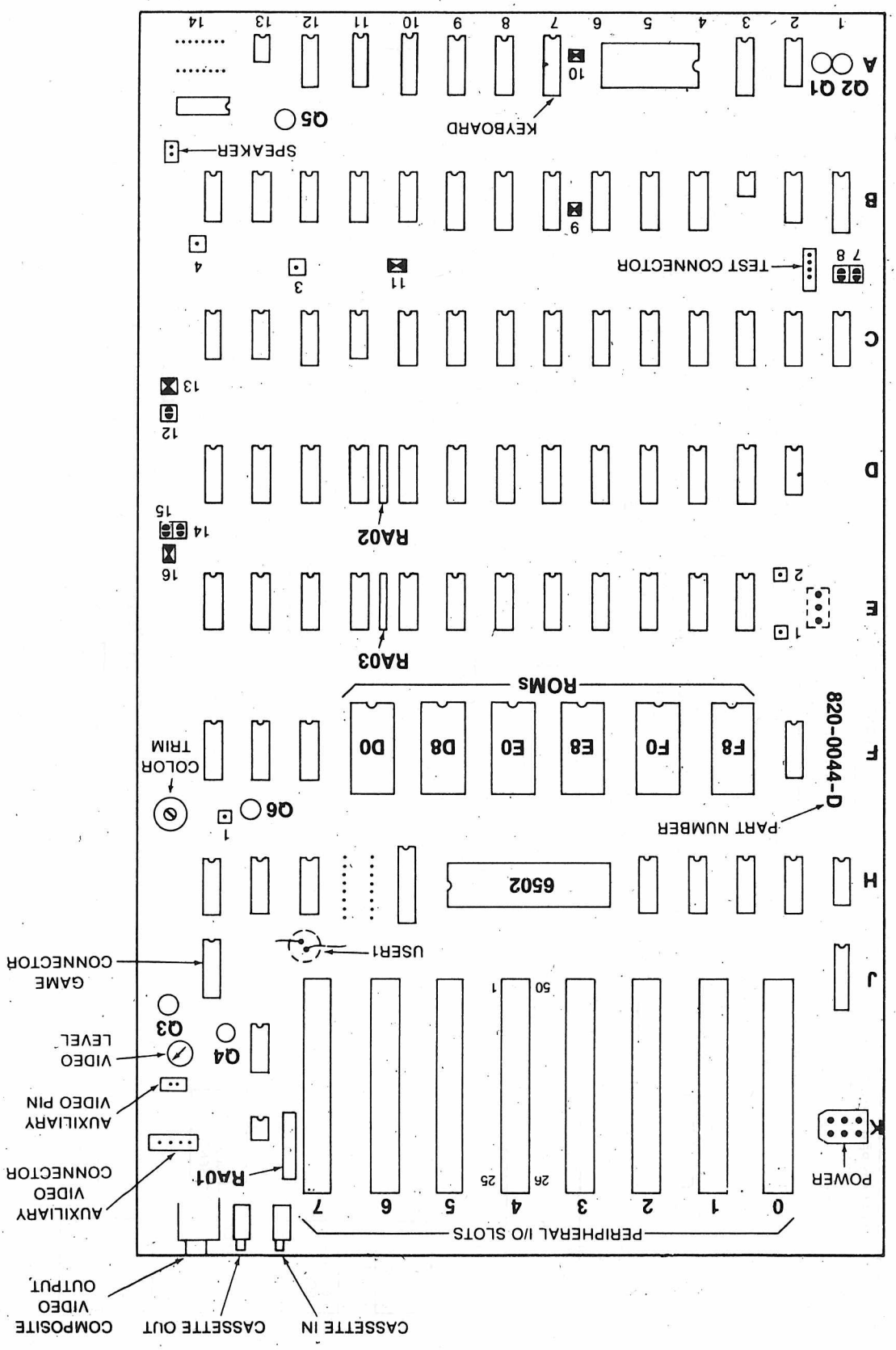



Fig. 1-1. Diagram of the Apple II RFI mother board. (Courtesy Apple Computer Inc.)

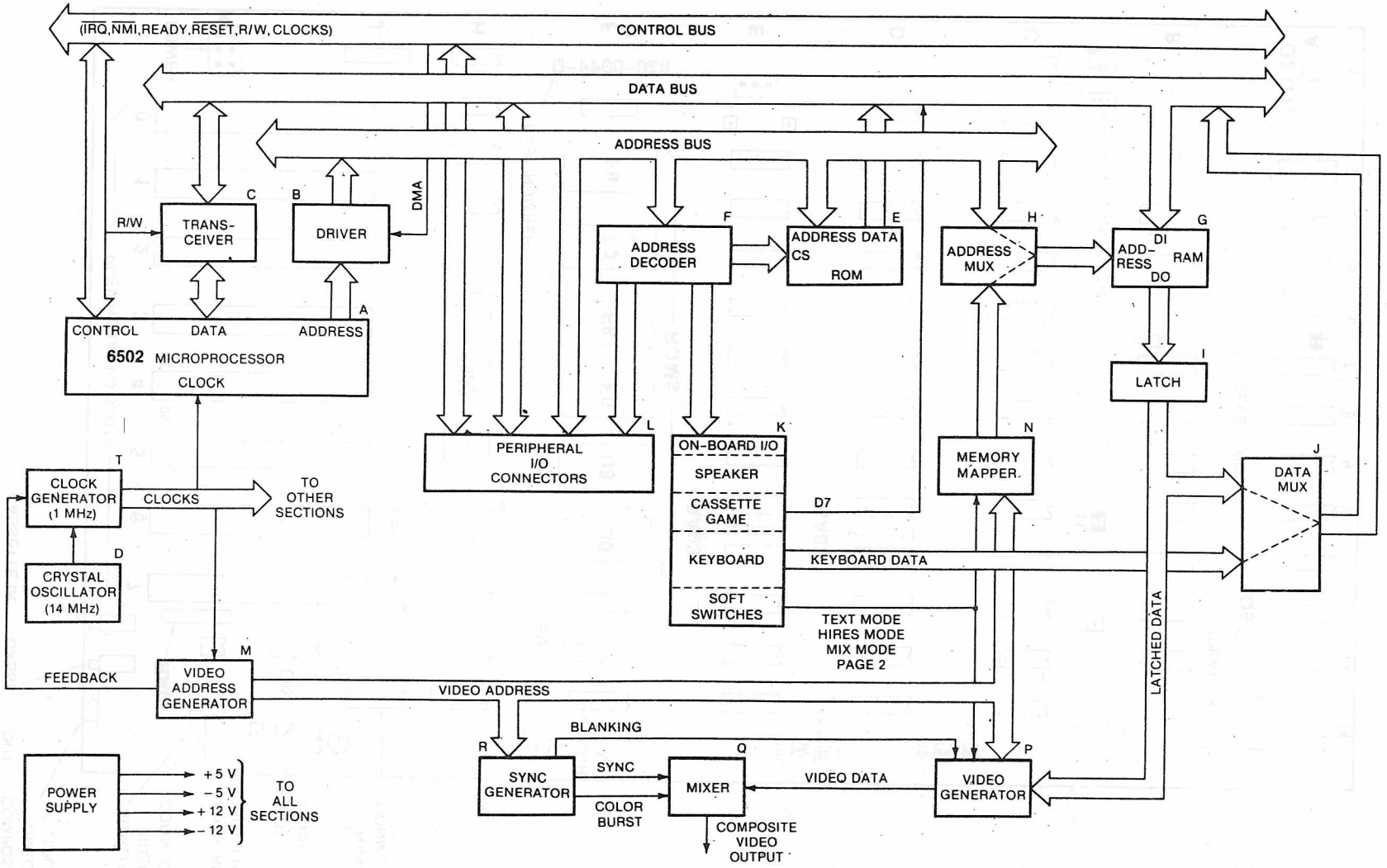



Fig. 2-1. Apple II block diagram.

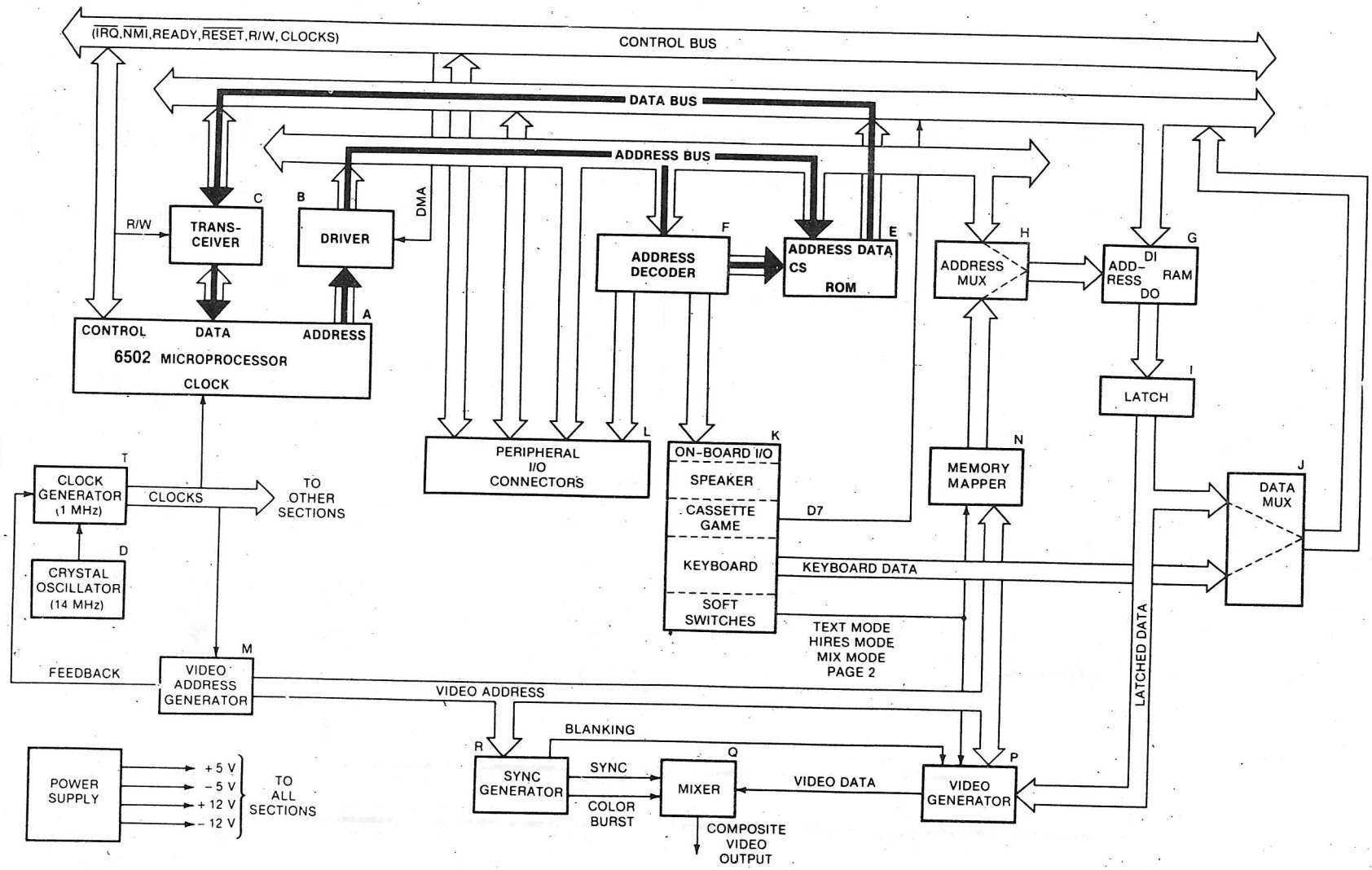



Fig. 2-2. ROM read-cycle path.

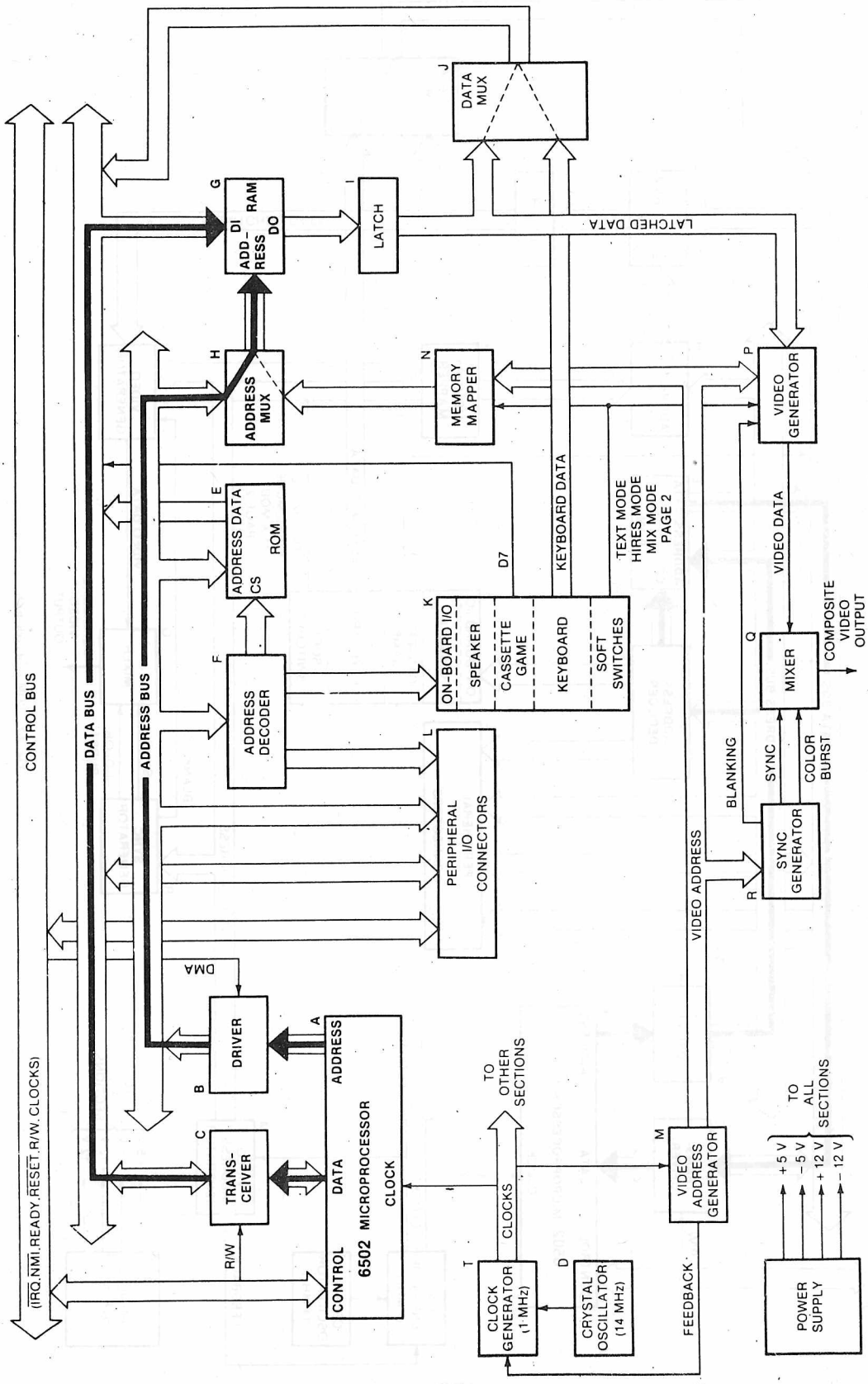



Fig. 2-3. RAM write-cycle path.

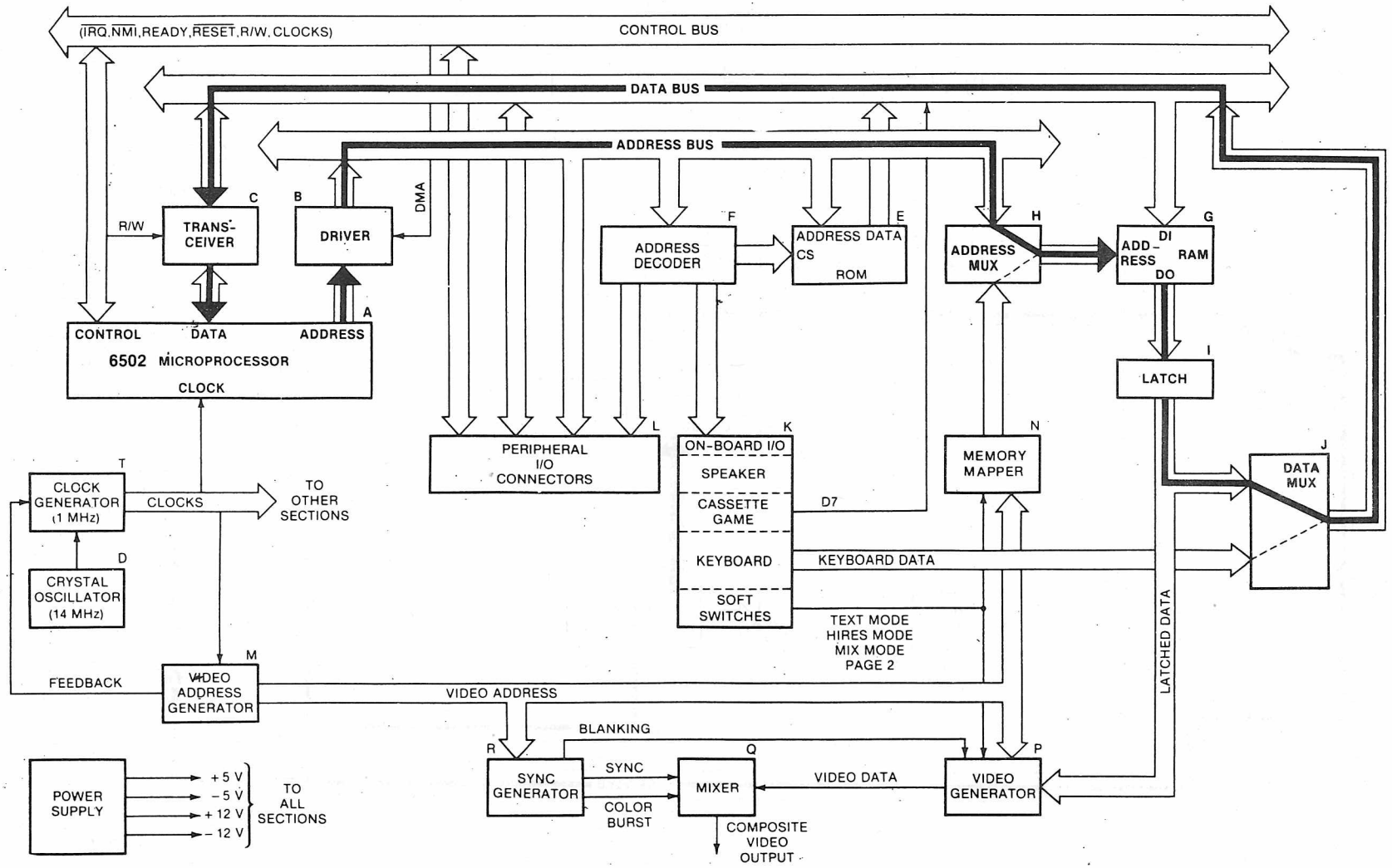



Fig. 2-4. RAM read-cycle path.

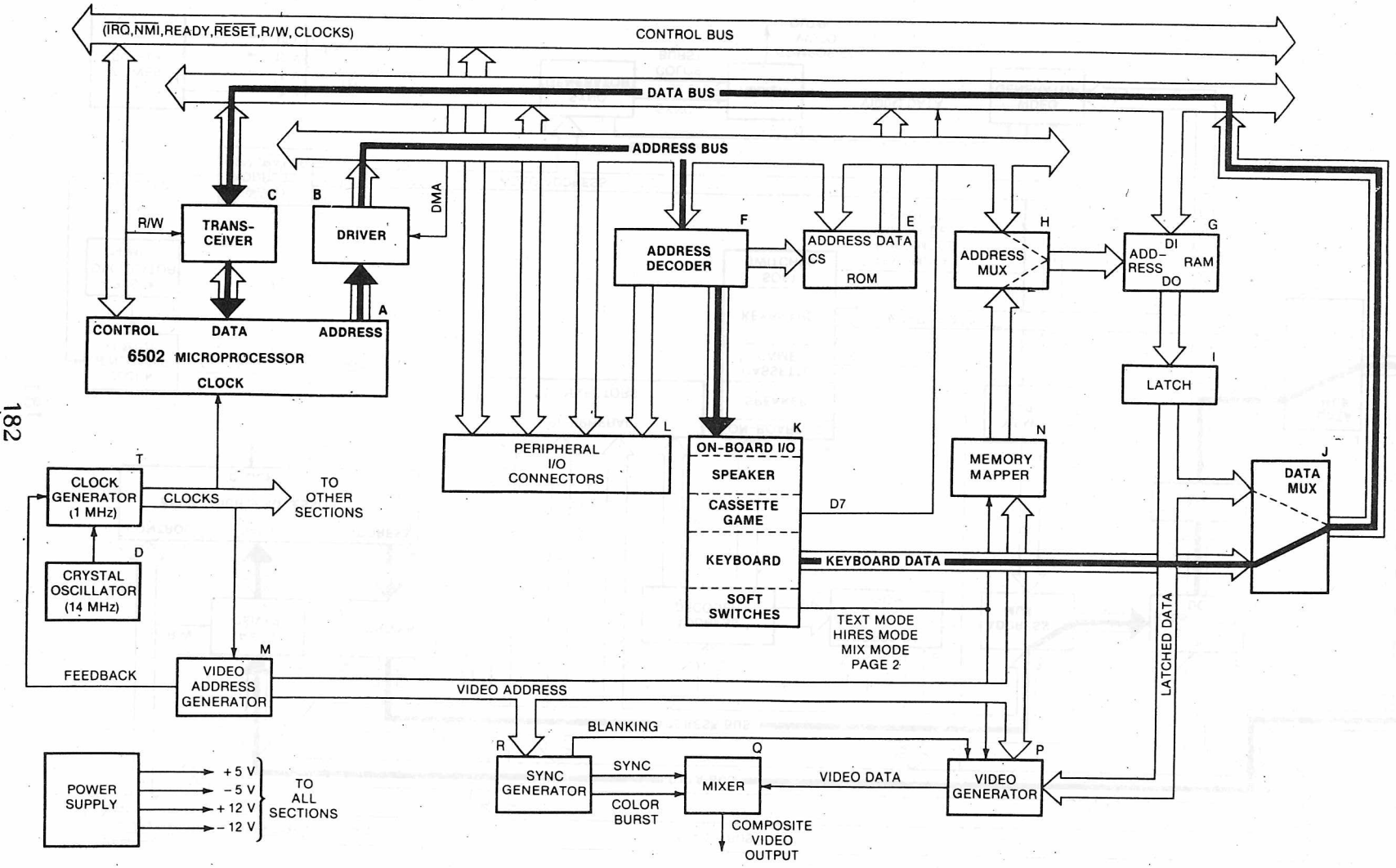



**Fig. 2-5.** Keyboard read-cycle path.

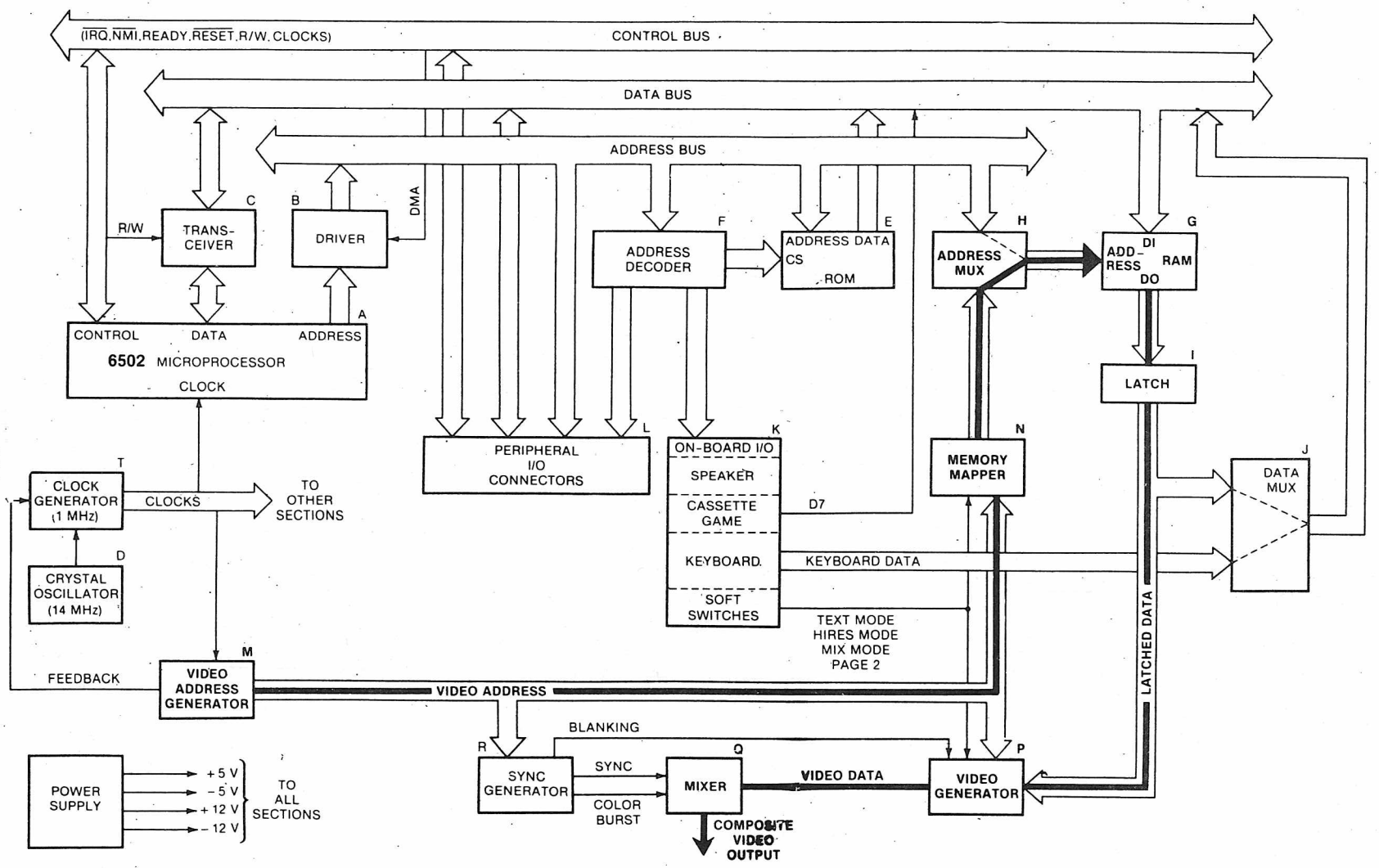
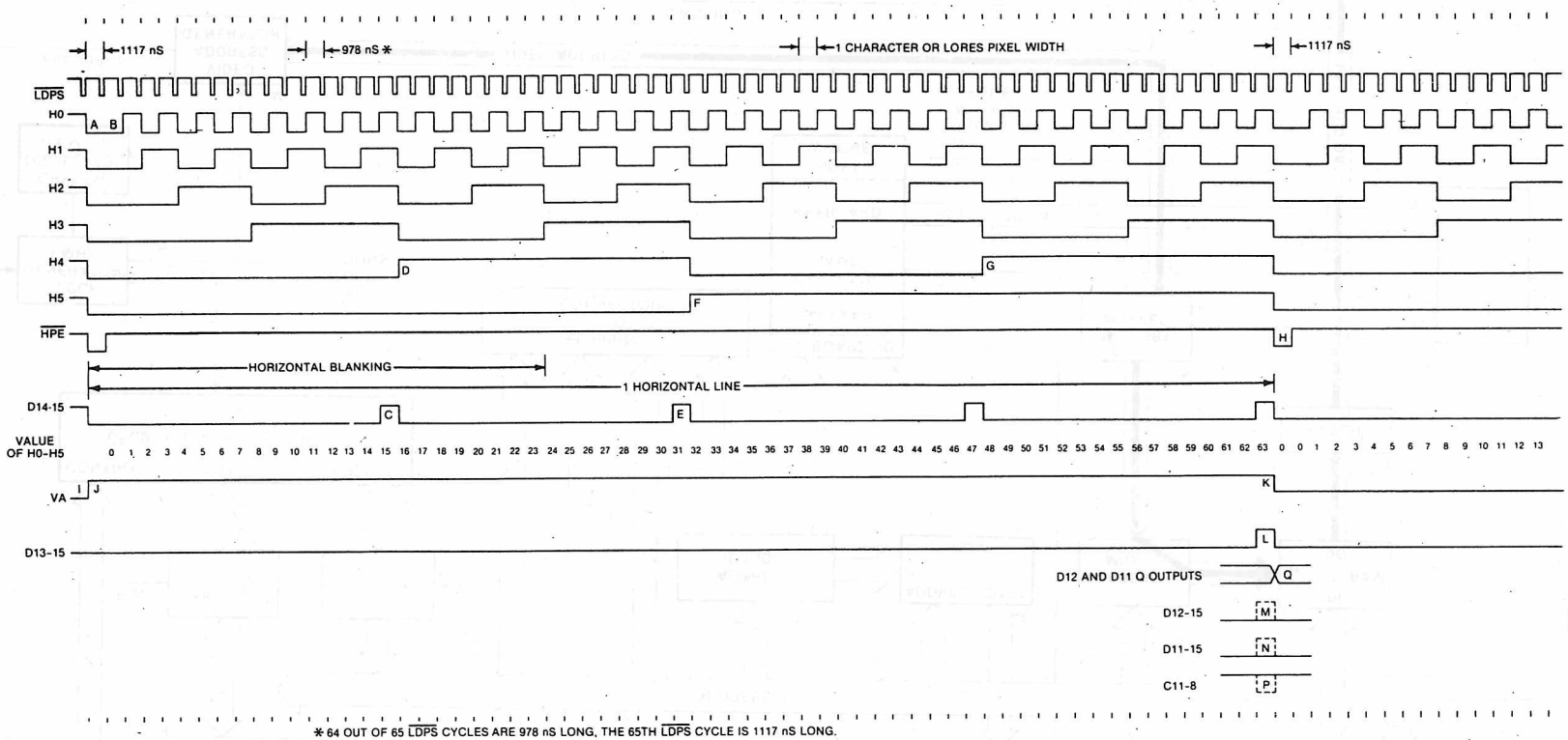



Fig. 2-6. Video-cycle path.




* 64 OUT OF 65 $\overline{\text{LDPS}}$ CYCLES ARE 978 nS LONG, THE 65TH $\overline{\text{LDPS}}$ CYCLE IS 1117 nS LONG.

Fig. 3-4. Horizontal timing.

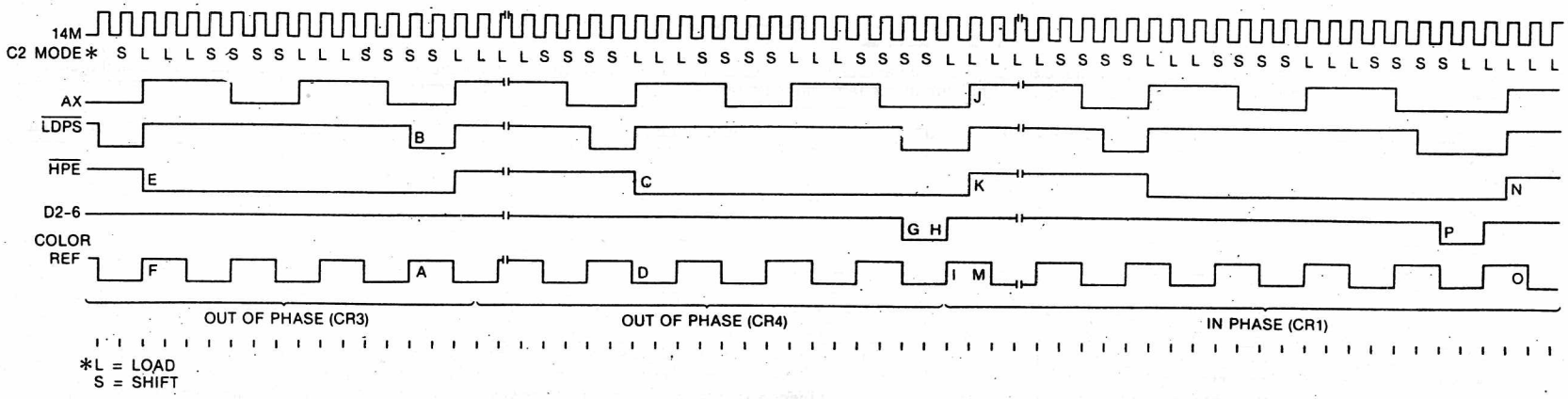



Fig. 3-7. Clock synchronization—CR3 and CR4.

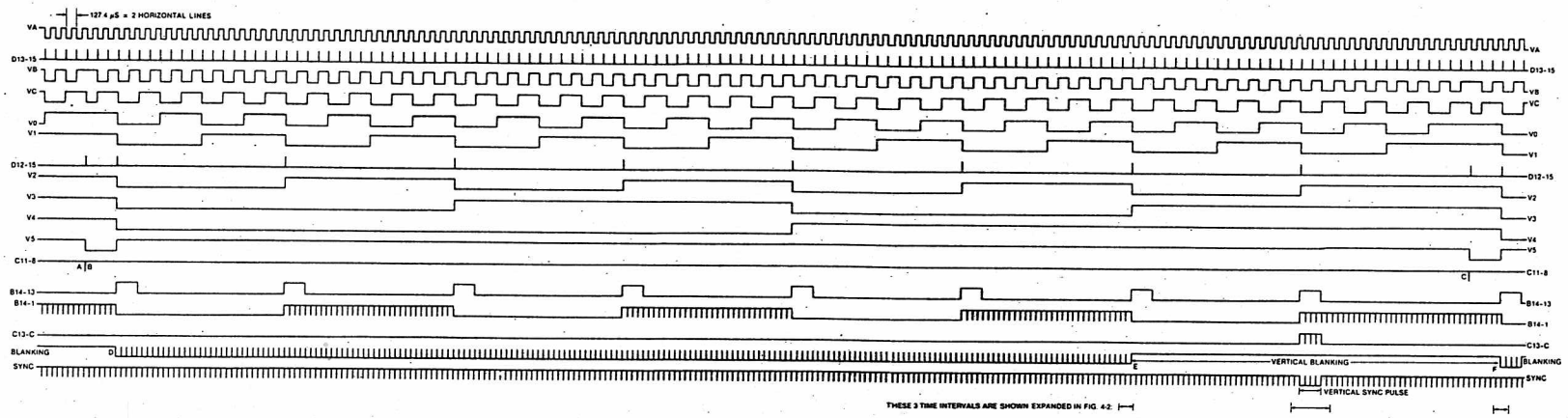



Fig. 4-1. Vertical timing.

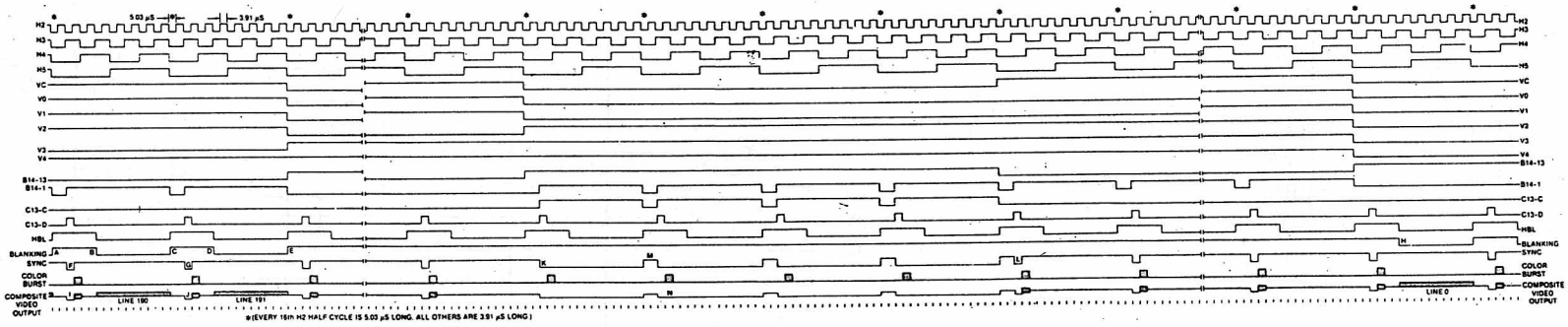



Fig. 4-2. Vertical retrace interval.

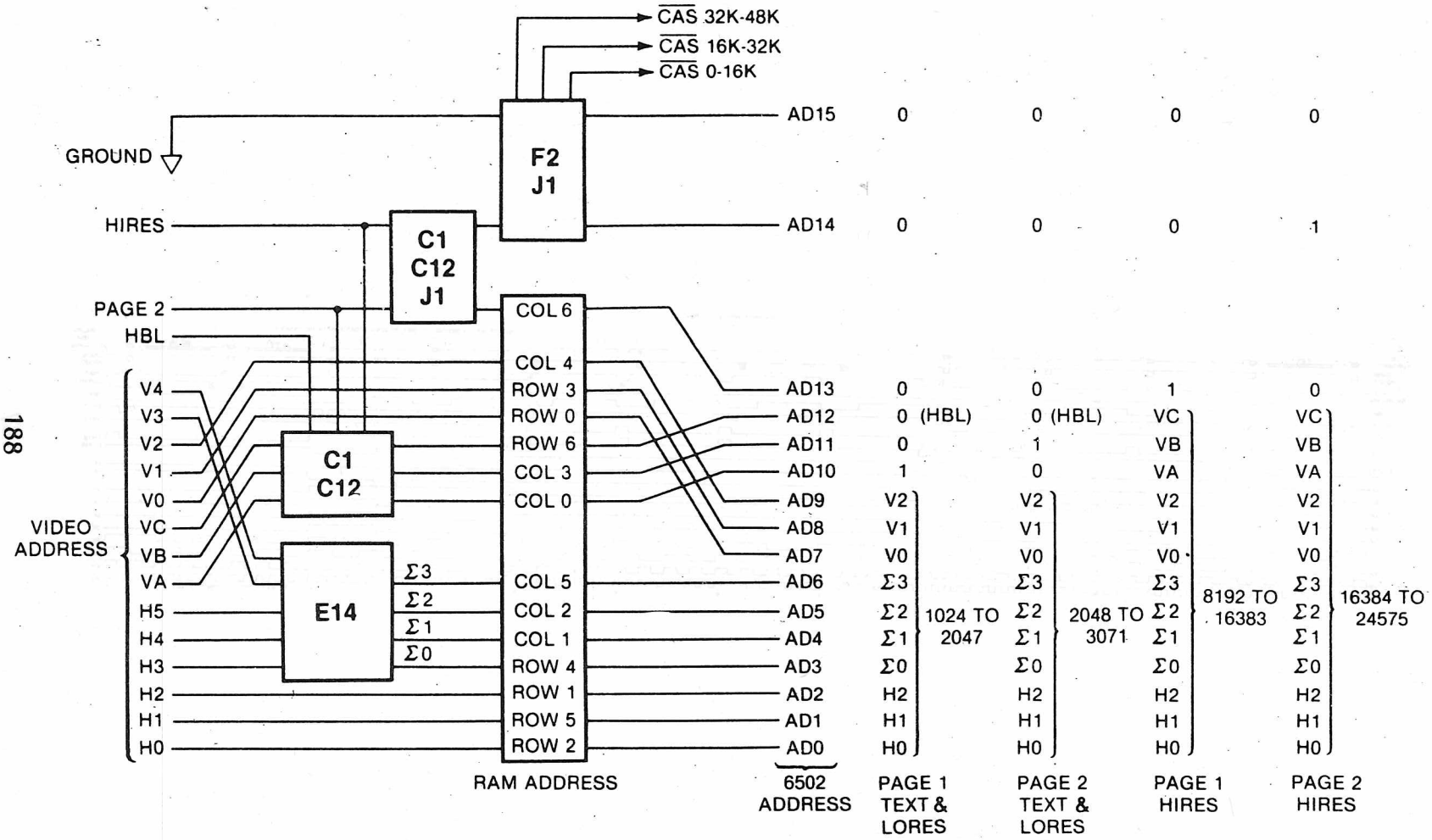
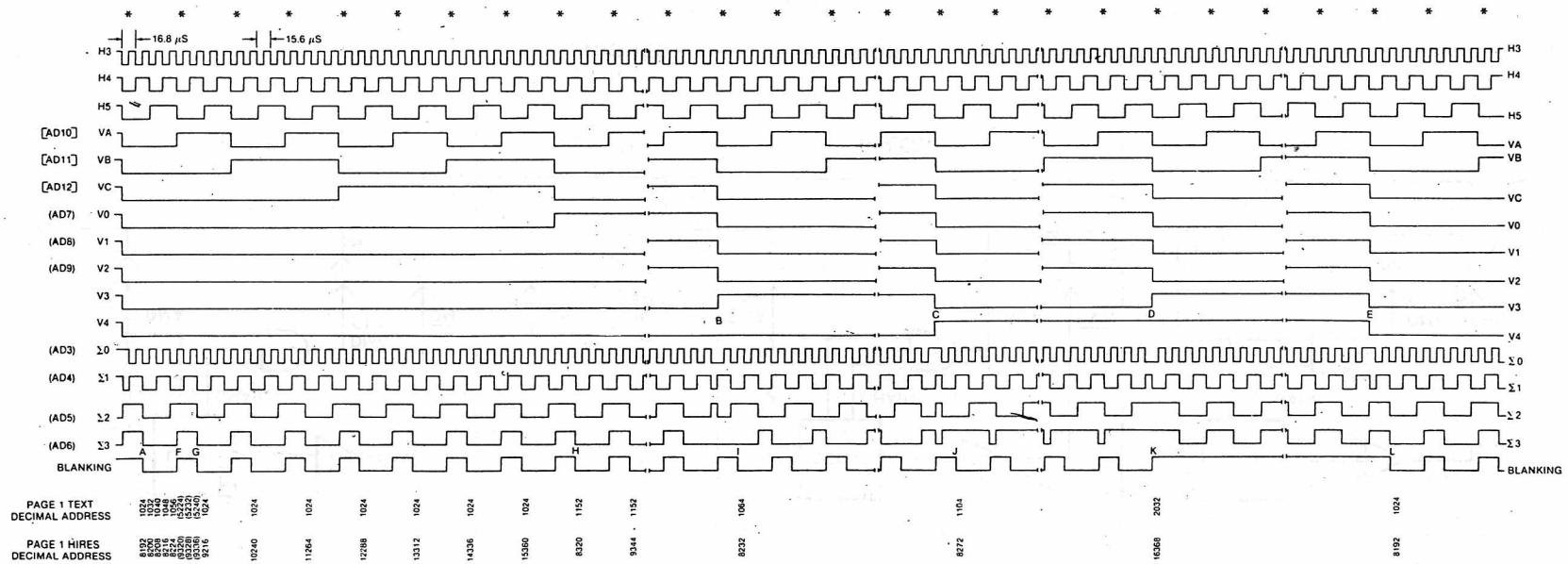



**Fig. 5-7.** Video mapping.




*EVERY FOURTH H3 ..PERIOD IS 16.8 μS LONG. ALL OTHERS ARE 15.6 μS LONG.

Fig. 5-8. Video address signals.

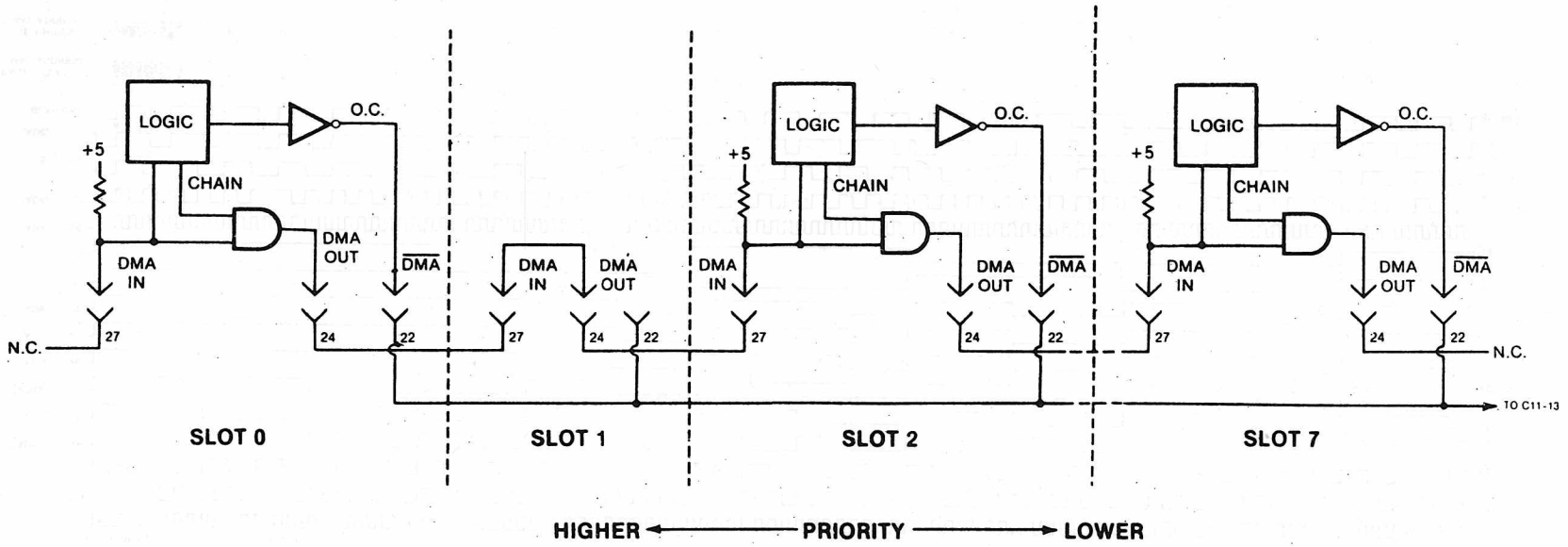



Fig. 6-11. DMA daisy chain.

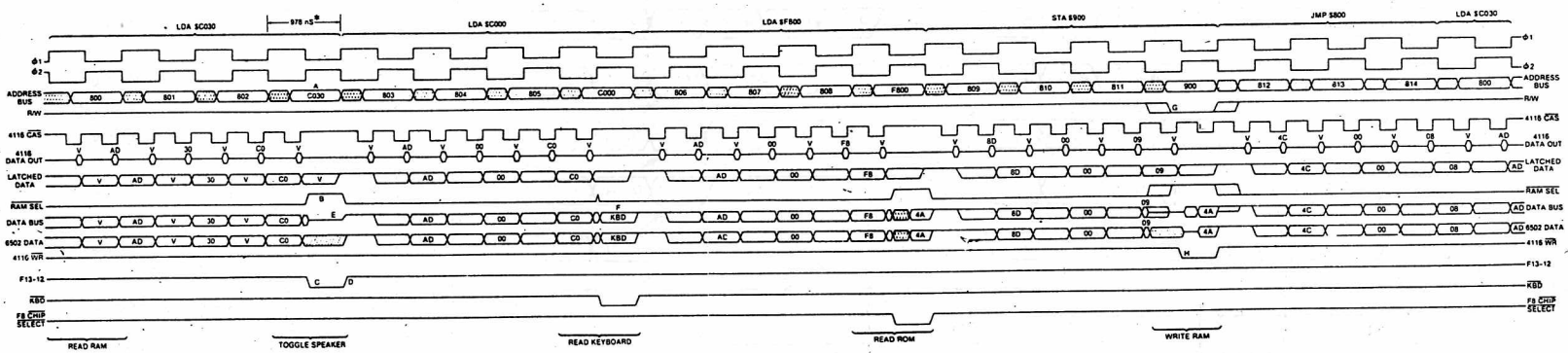



Fig. 6-16. 6502 scope loop.

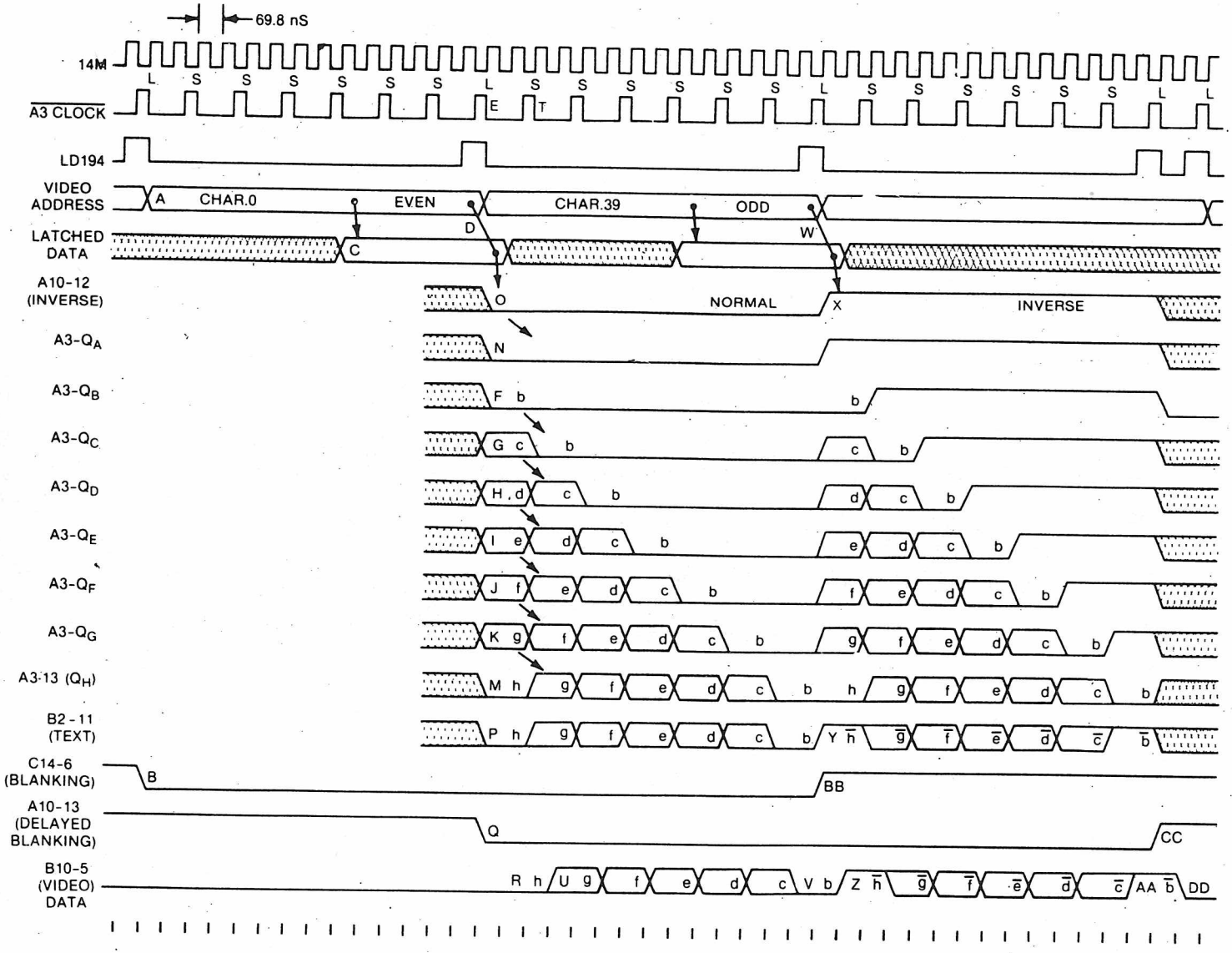



**Fig. 8-26.** Text mode timing diagram.

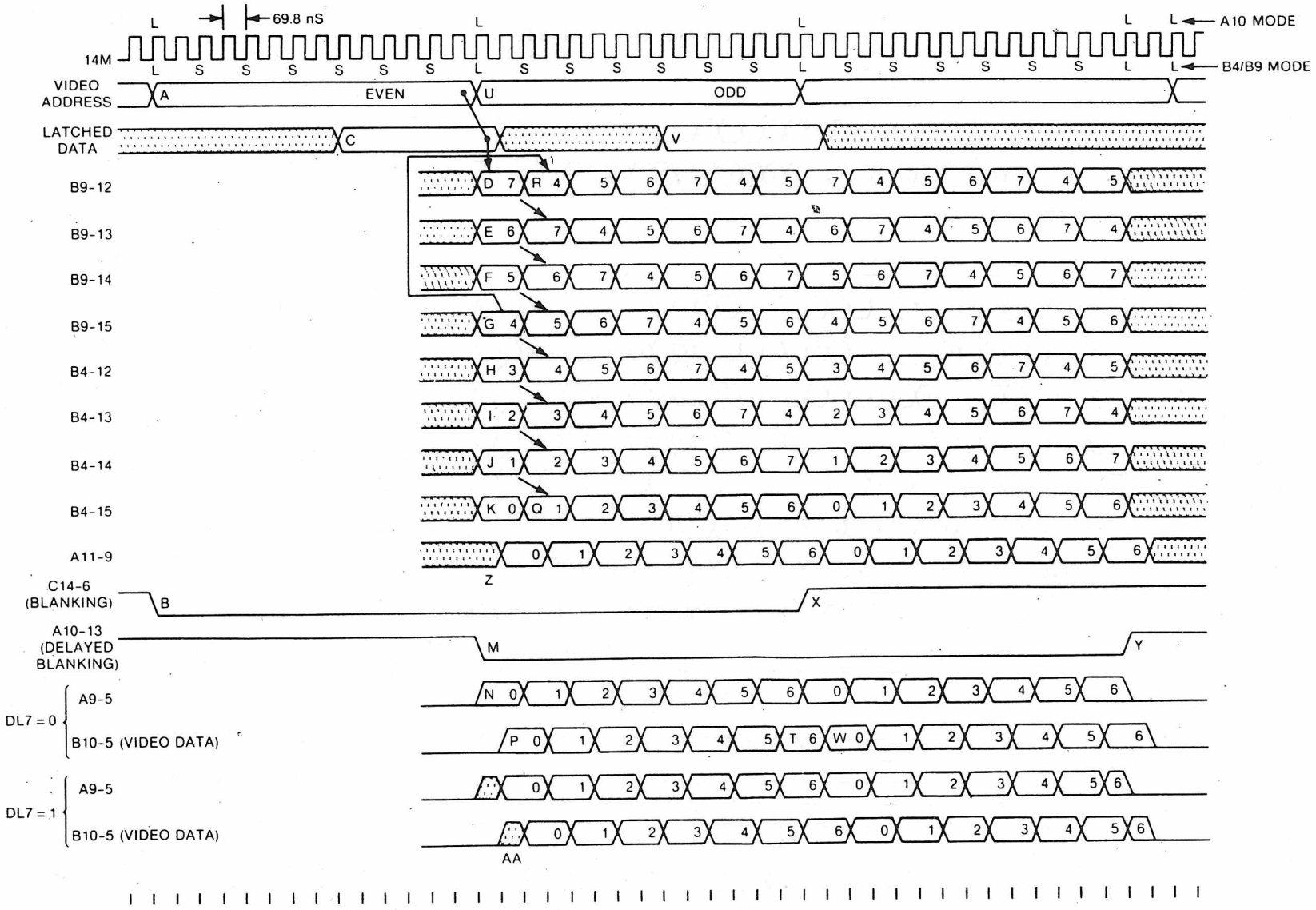



**Fig. 8-28.** HIRES-mode timing diagram.

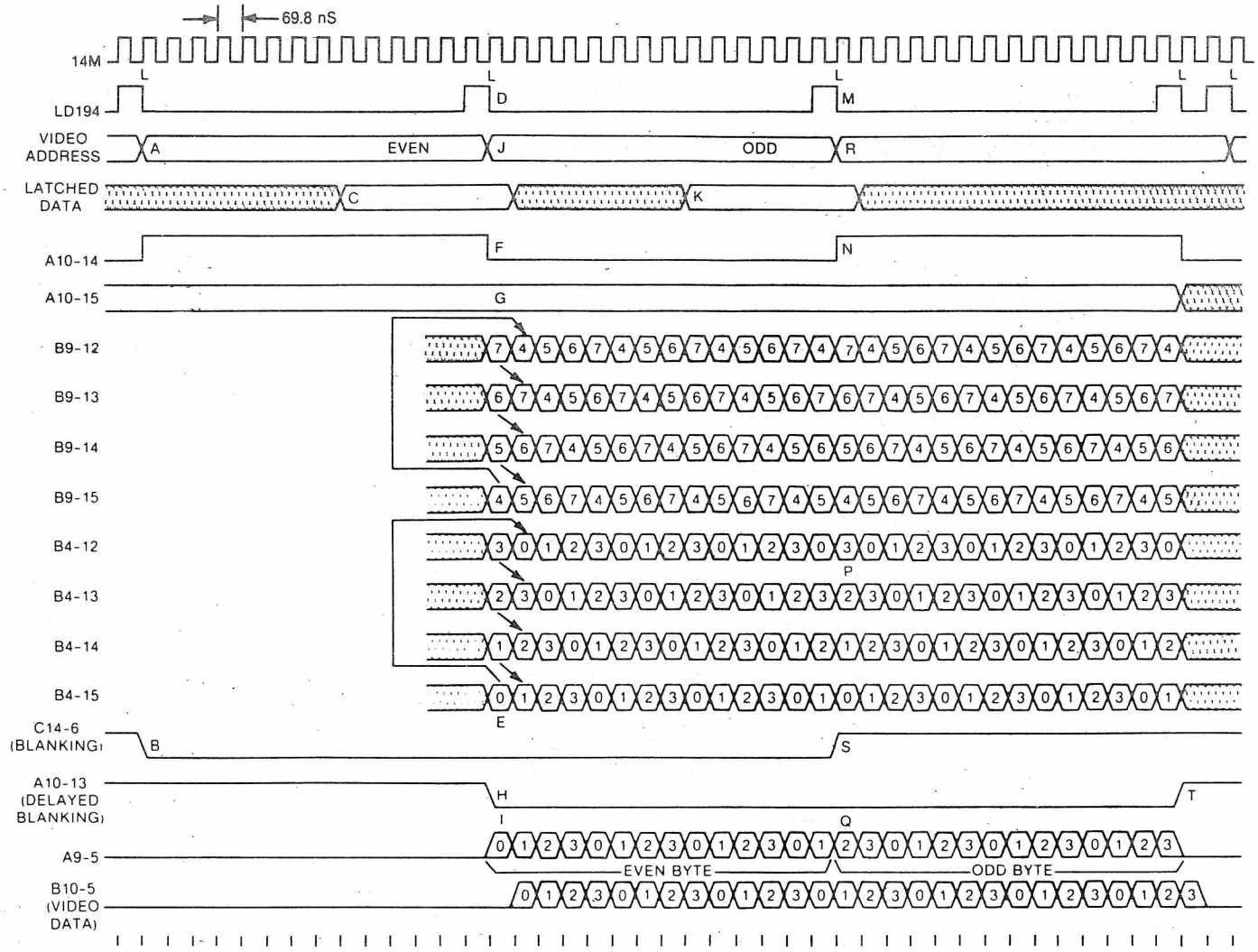



**Fig. 8-31.** LORES-mode timing diagram.

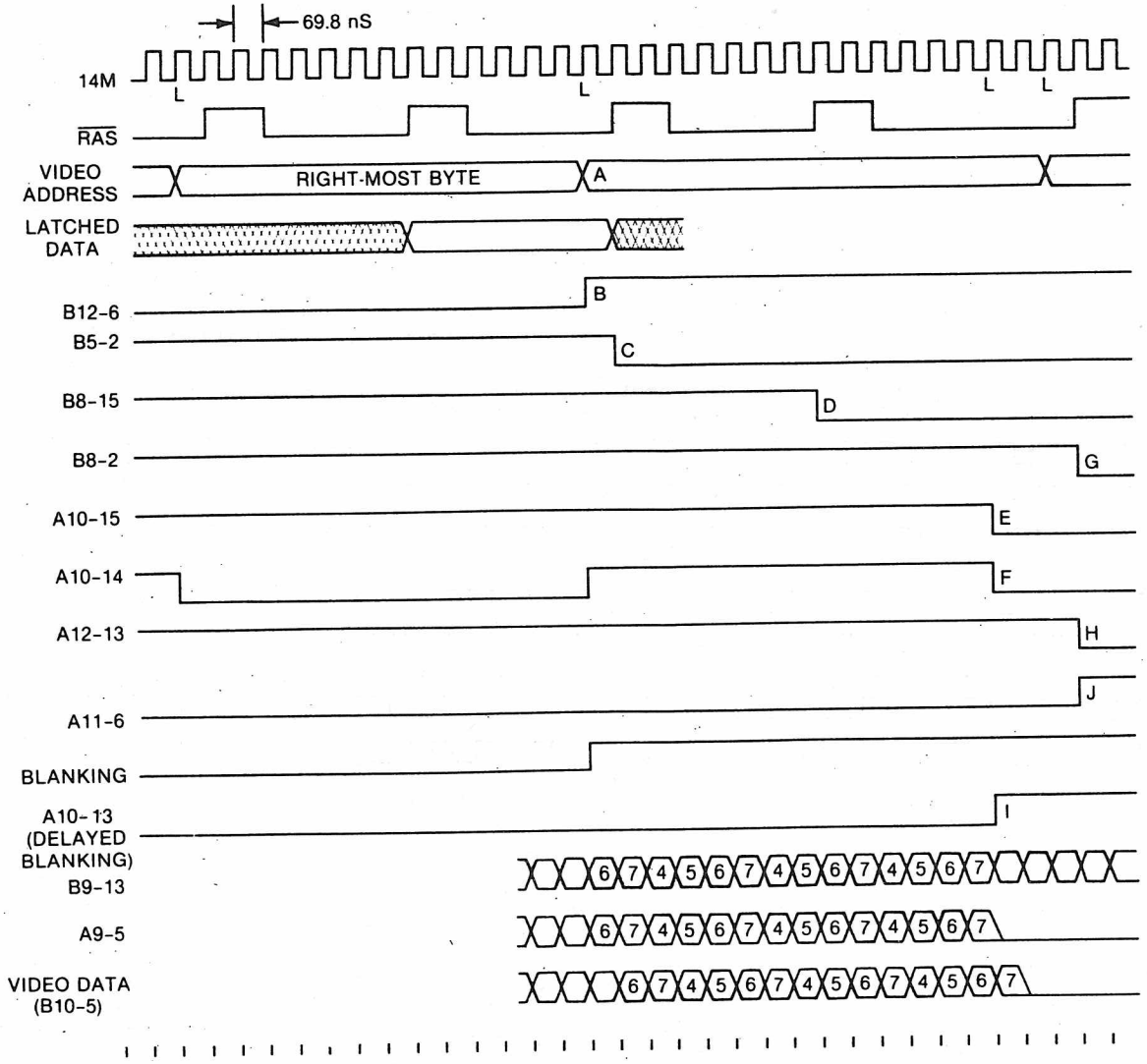



**Fig. 8-34.** Mixed LORES and text.

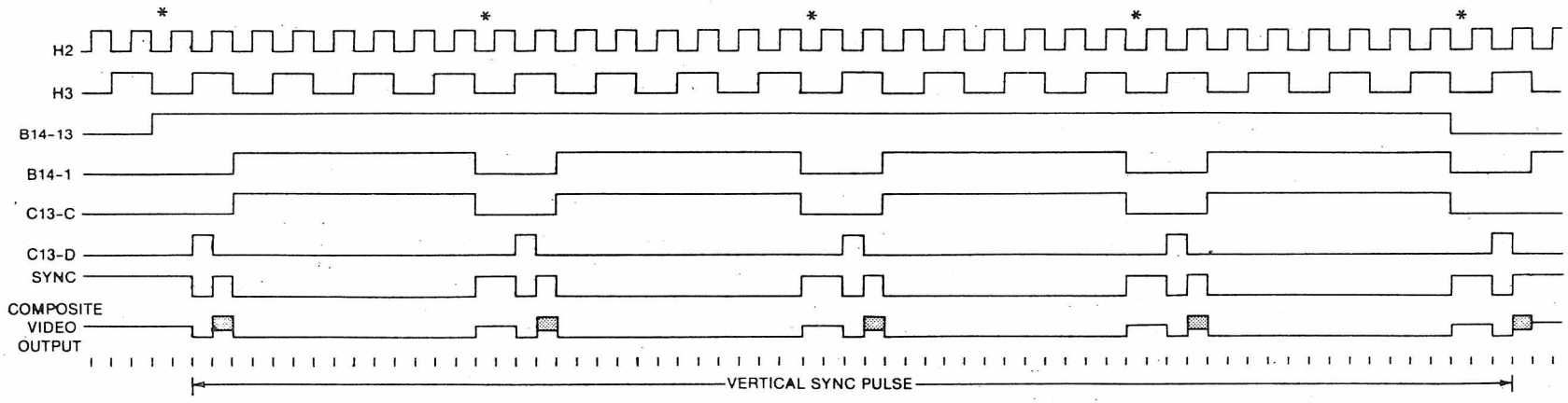



Fig. B-2. Vertical sync—Rev. 7.

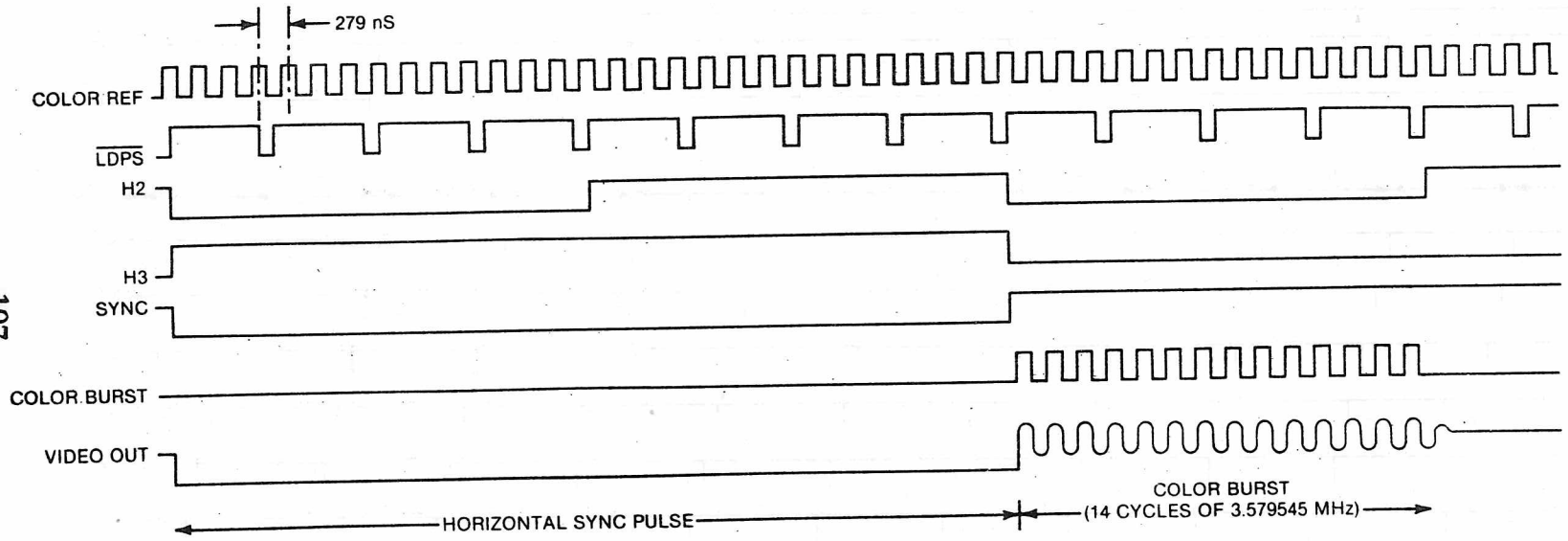



**Fig. B-4.** Color burst—Rev. 0 and 1.

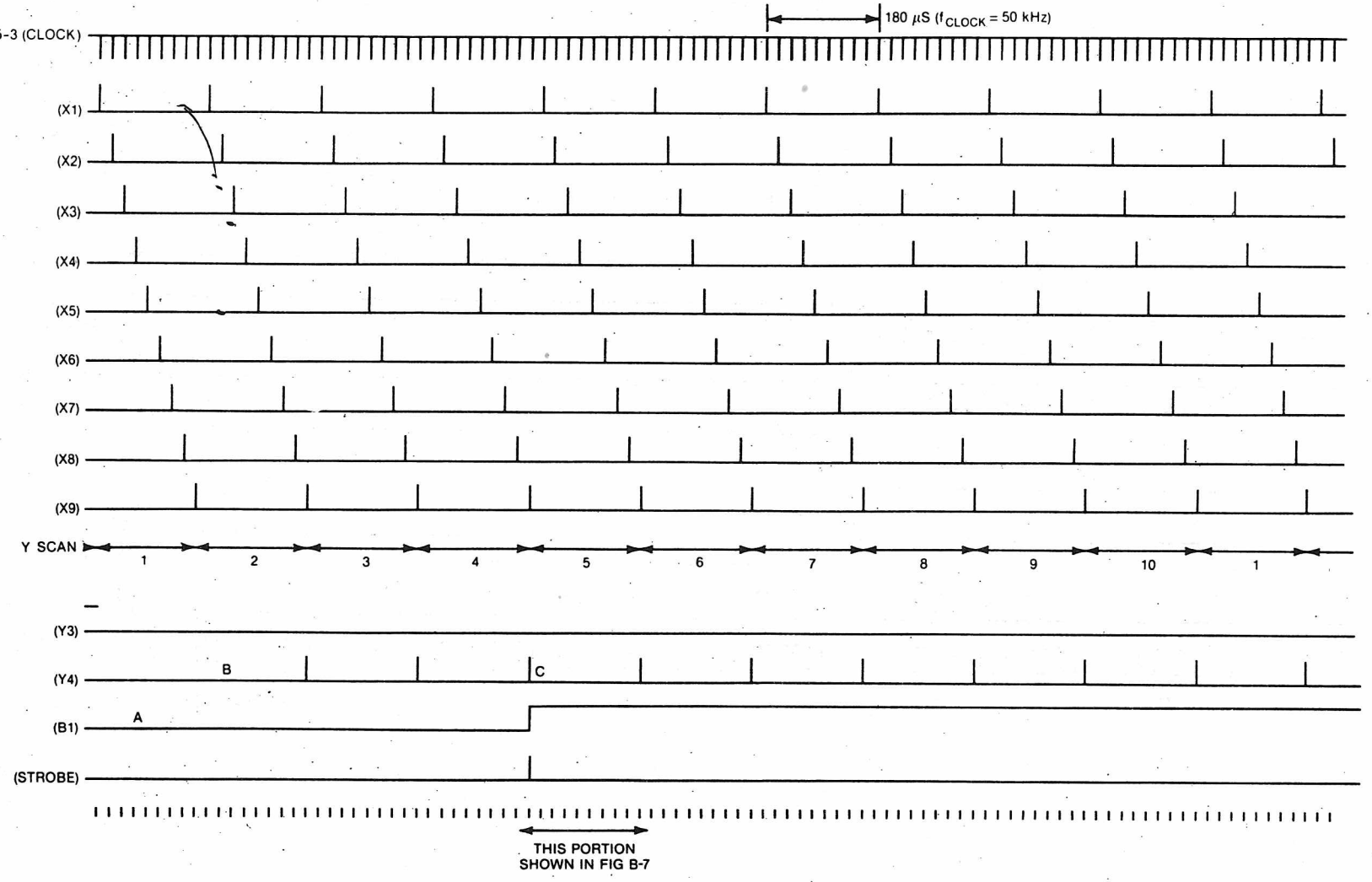



**Fig. B-8.** Y scan—single-piece keyboard.

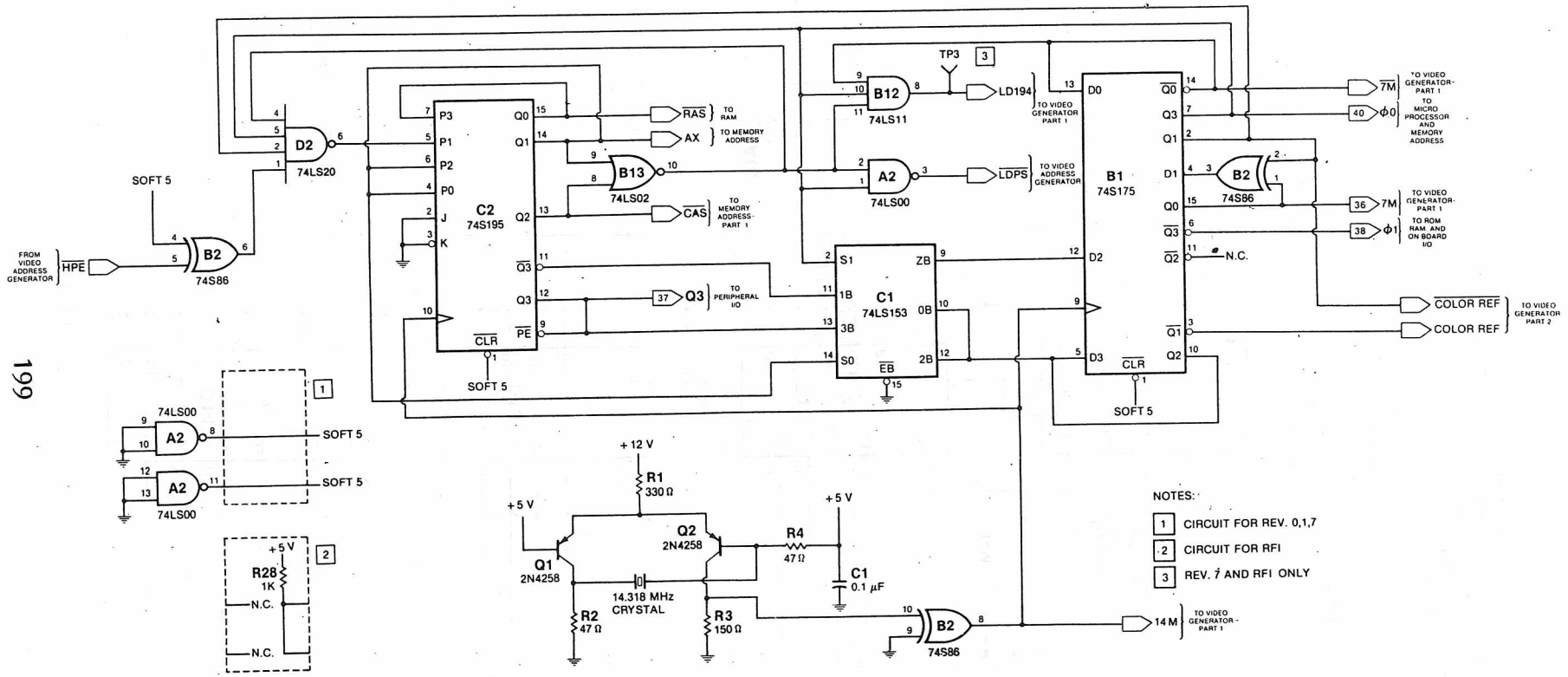



Fig. C-2. Clock generator (all revisions).

Fig. C-3. Video address generator (all revisions).

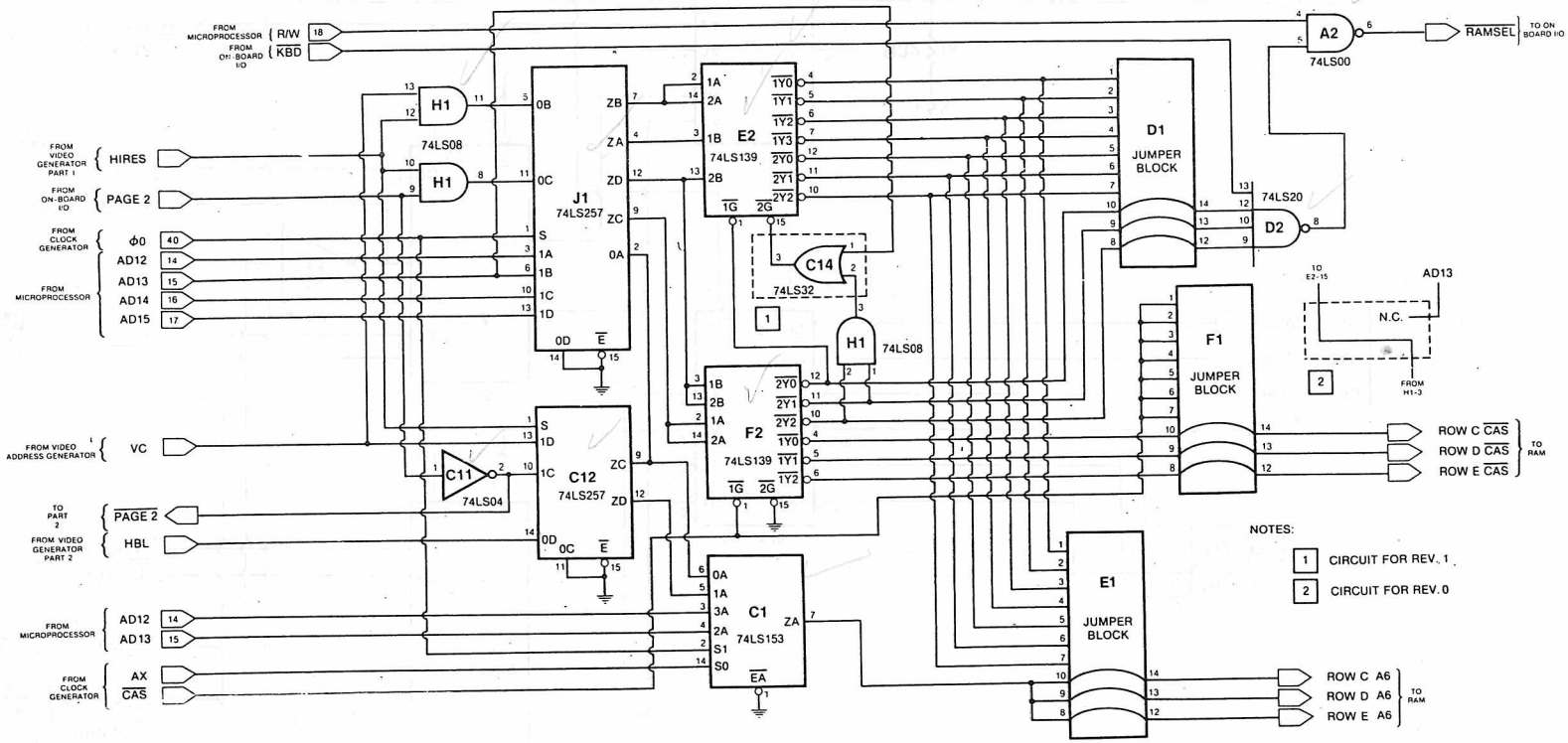



Fig. C-4. Memory address—part 1 (Rev. 0,1).

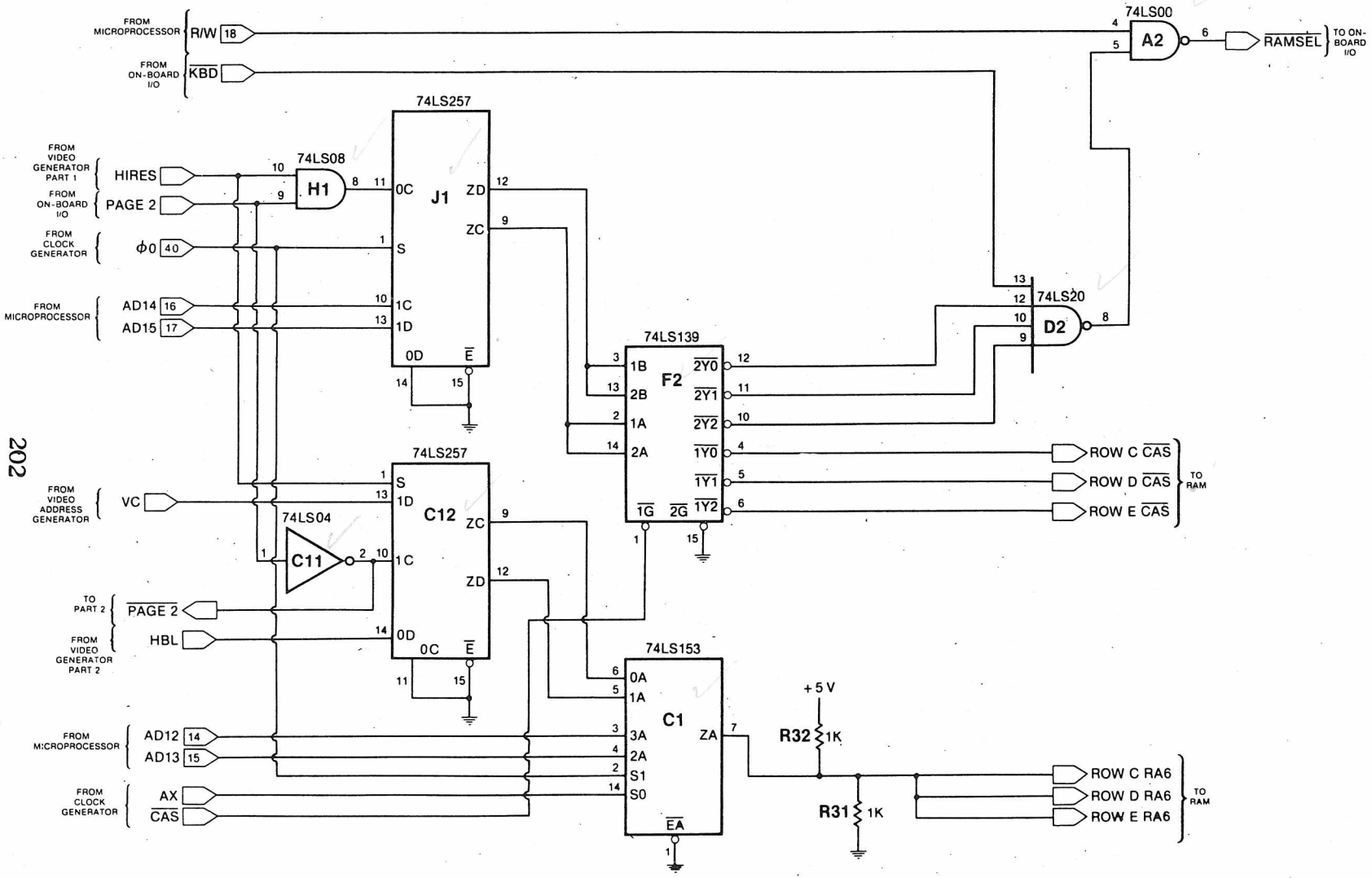



Fig. C-5. Memory address—part 1 (Rev. 7, RFI).

Fig. C-6. Memory address—part 2 (all revisions).

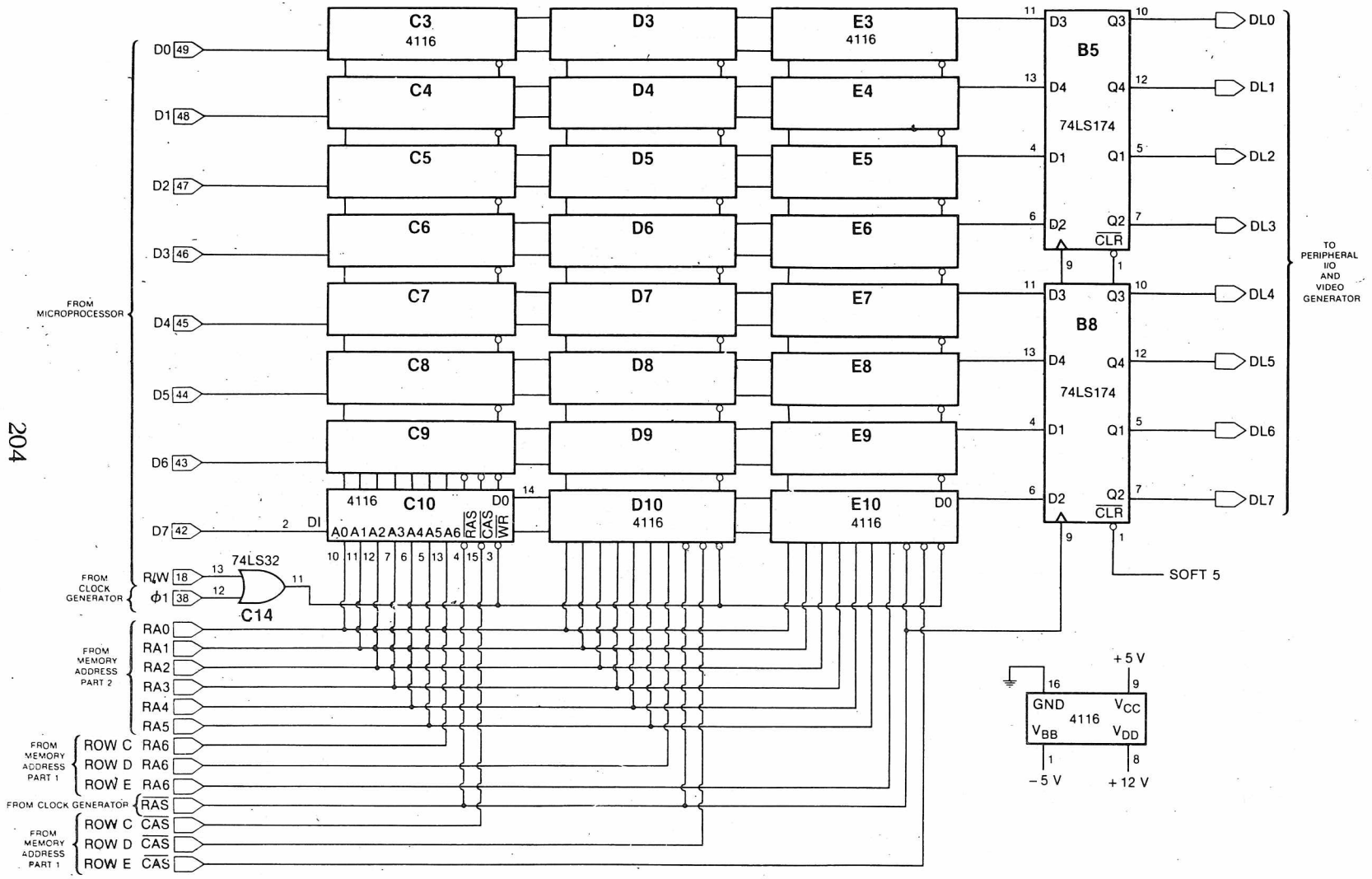



**Fig. C-7.** RAM (all revisions).

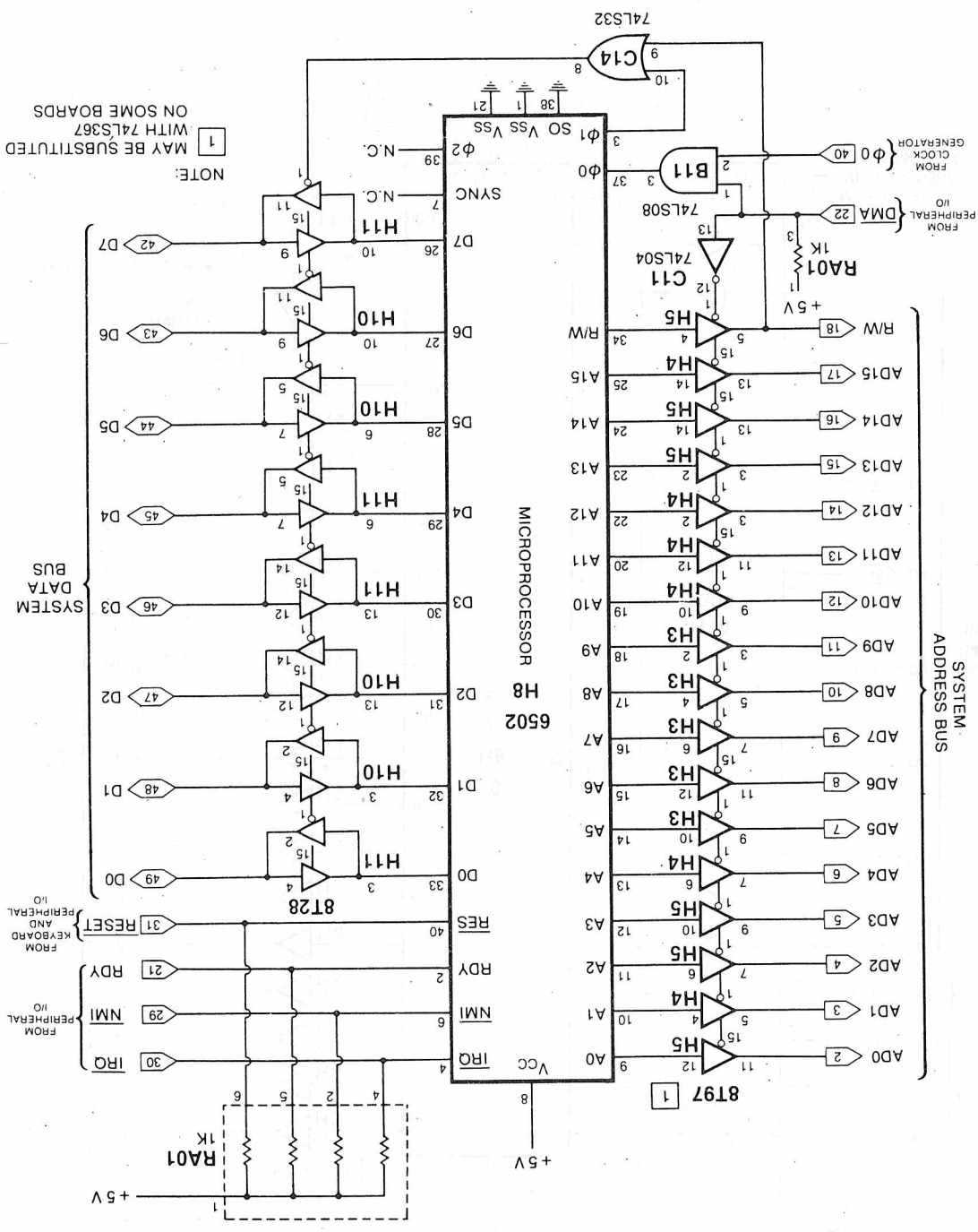



Fig. C-8. Microprocessor (Rev. 0.1,7).

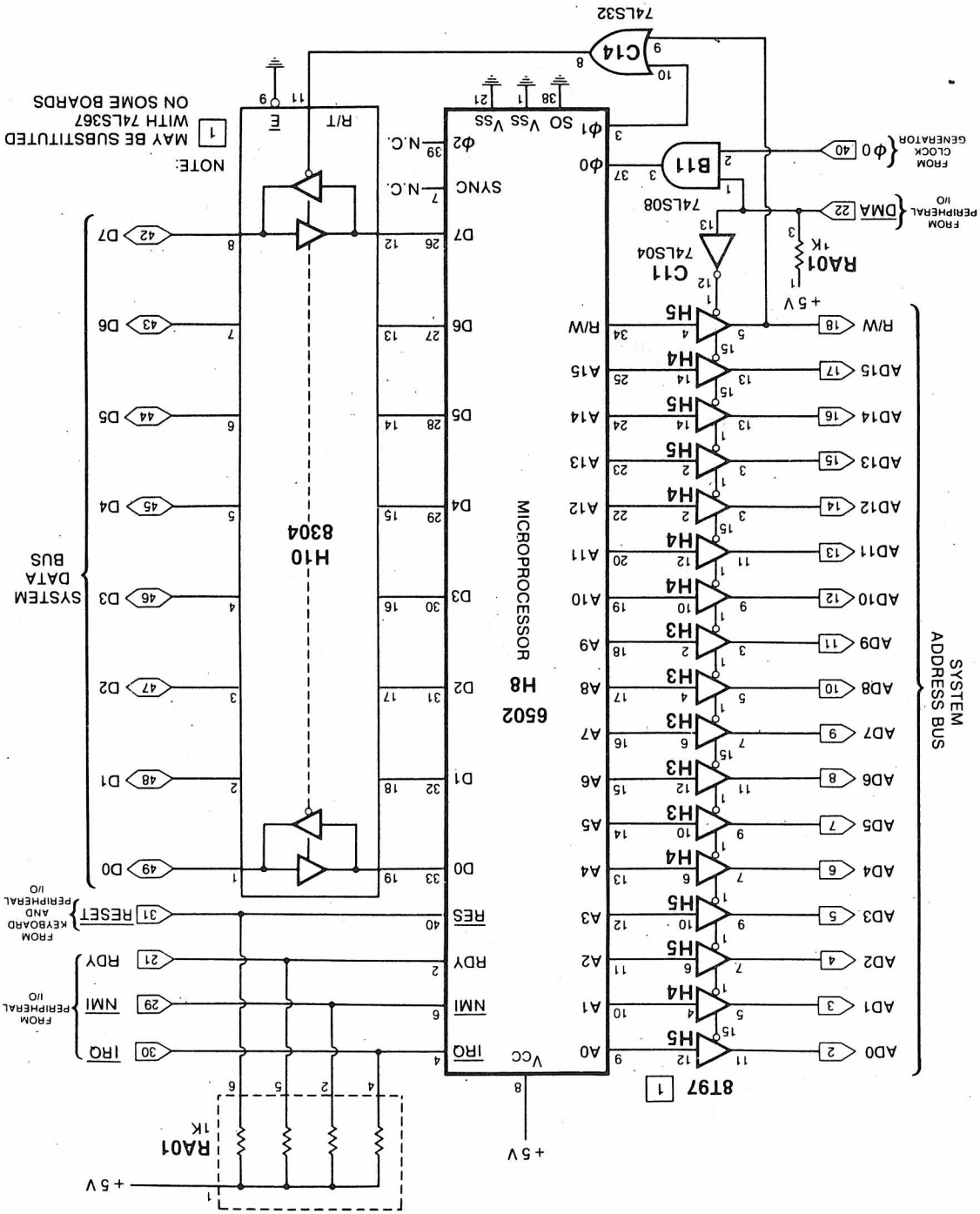



Fig. C-9. Microprocessor (RFI).

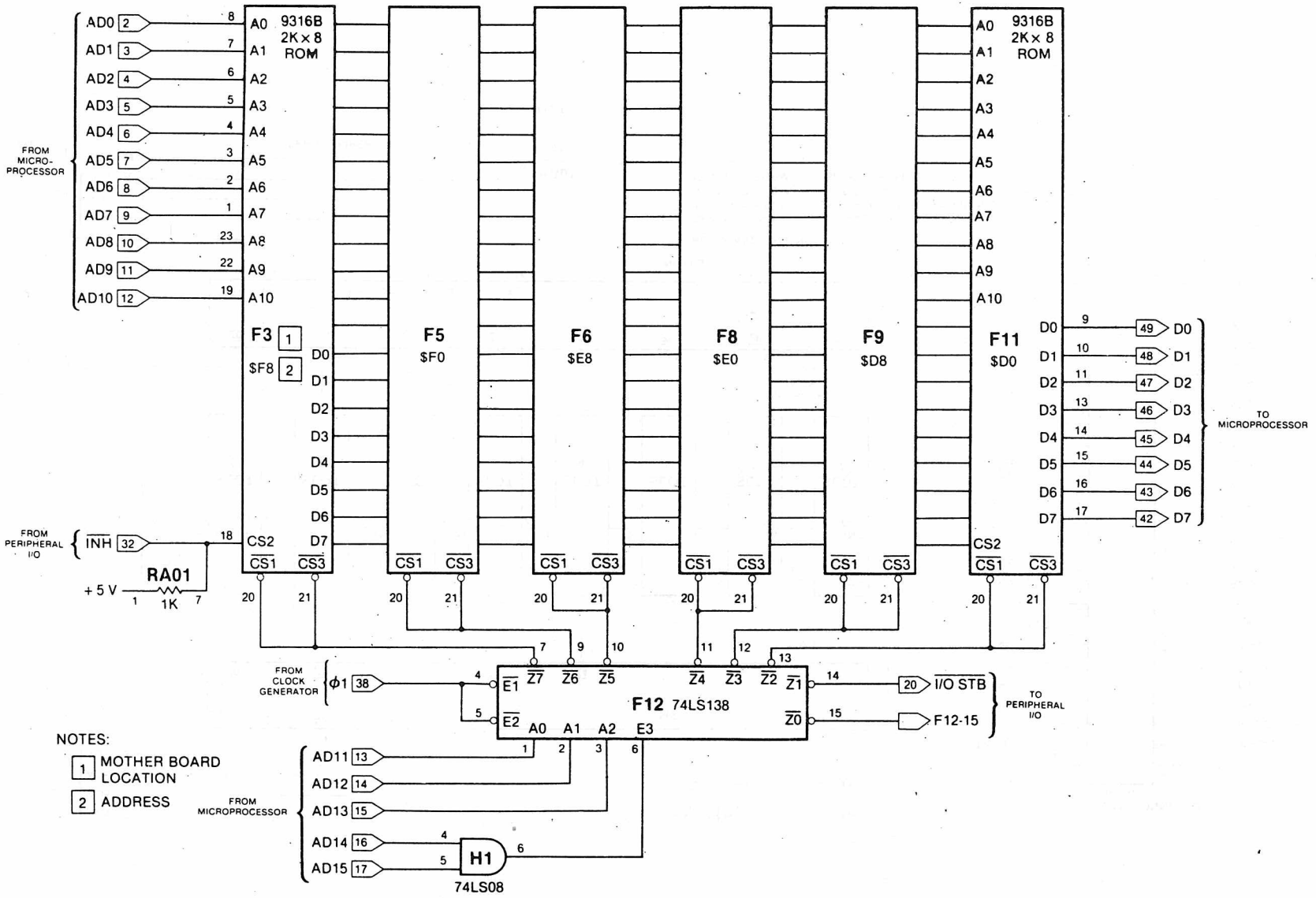



Fig. C-10. ROM (all revisions).

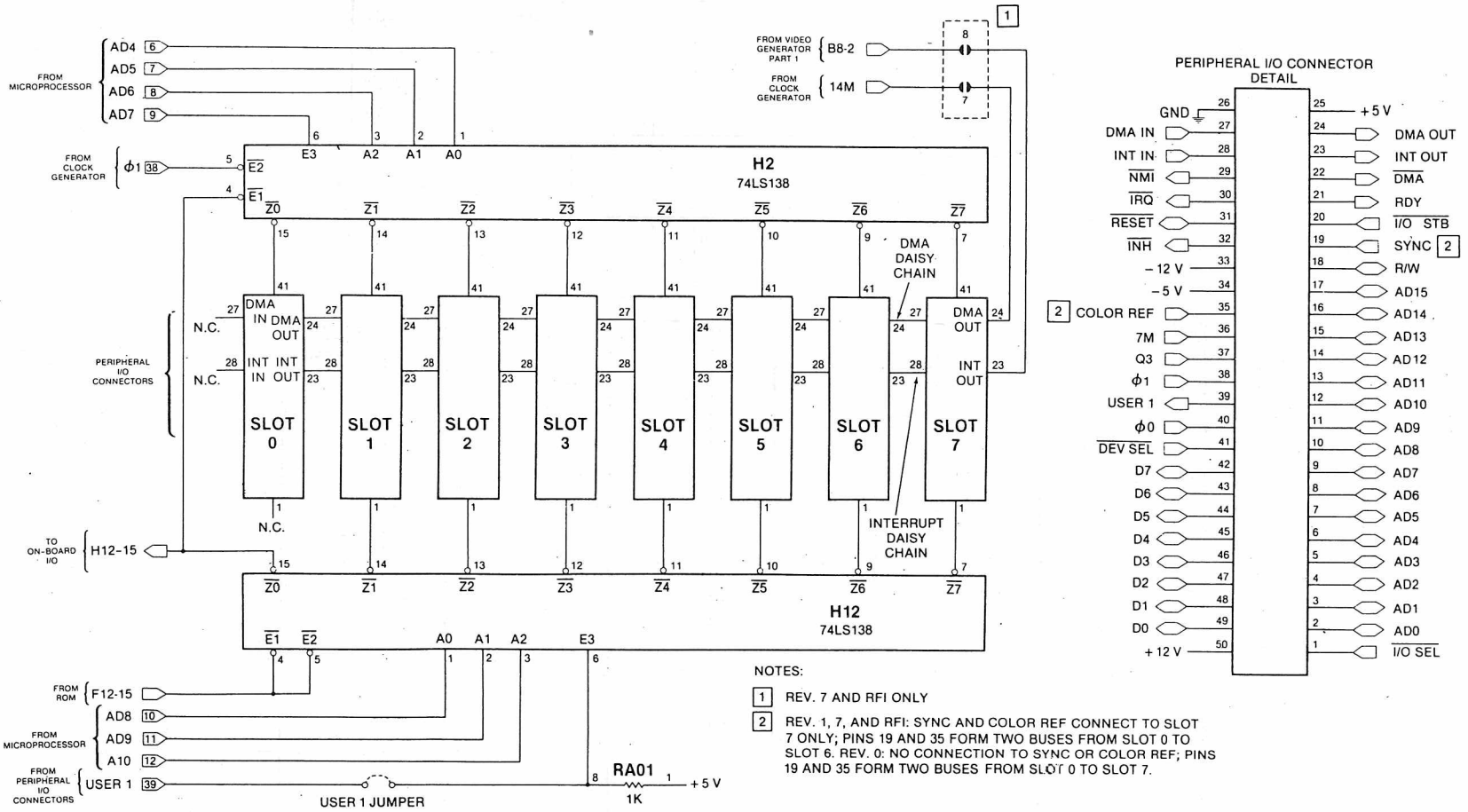



Fig. C-11. Peripheral I/O (all revisions).

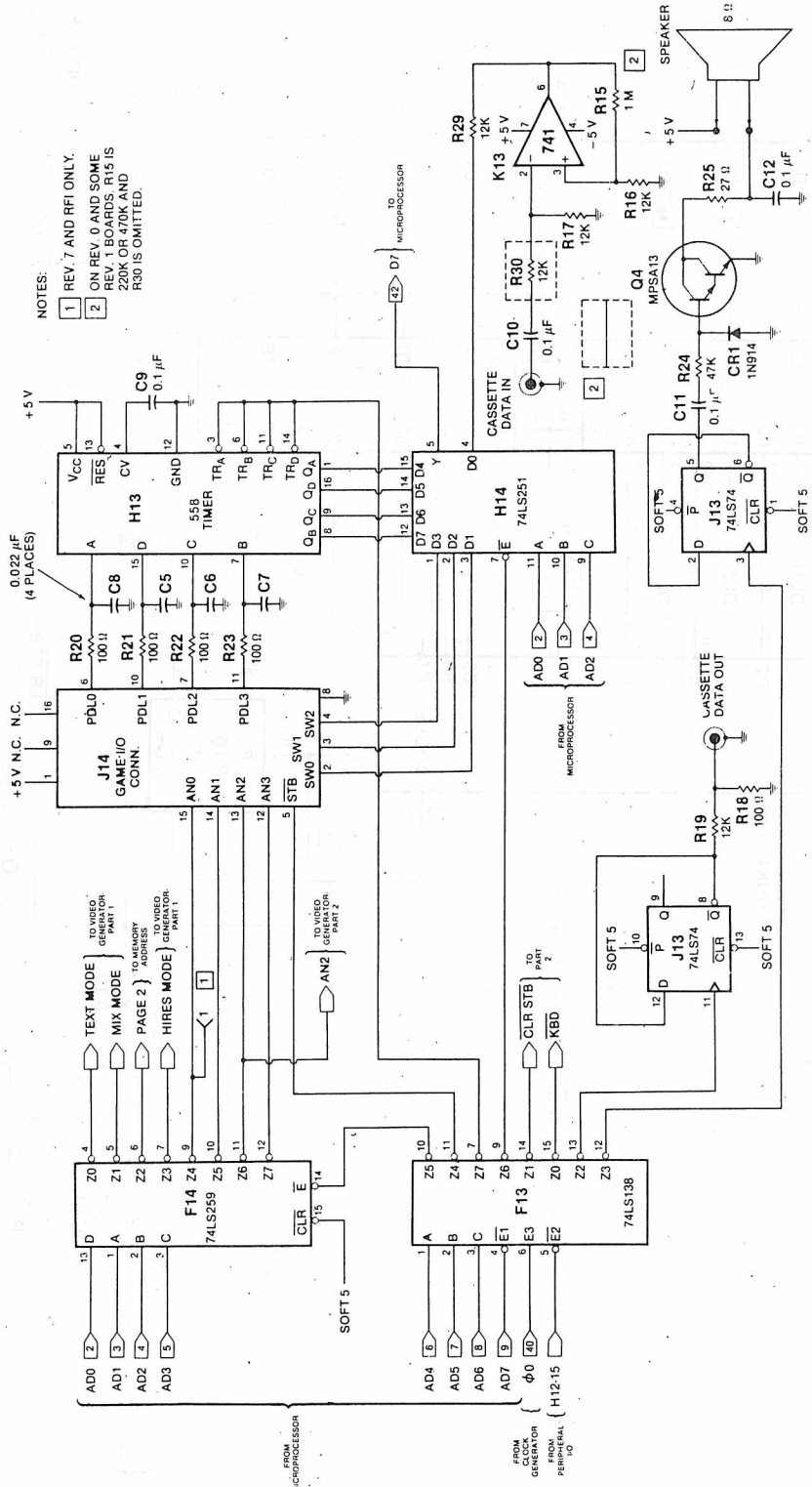



Fig. C-12. On-board I/O—part 1 (all revisions).

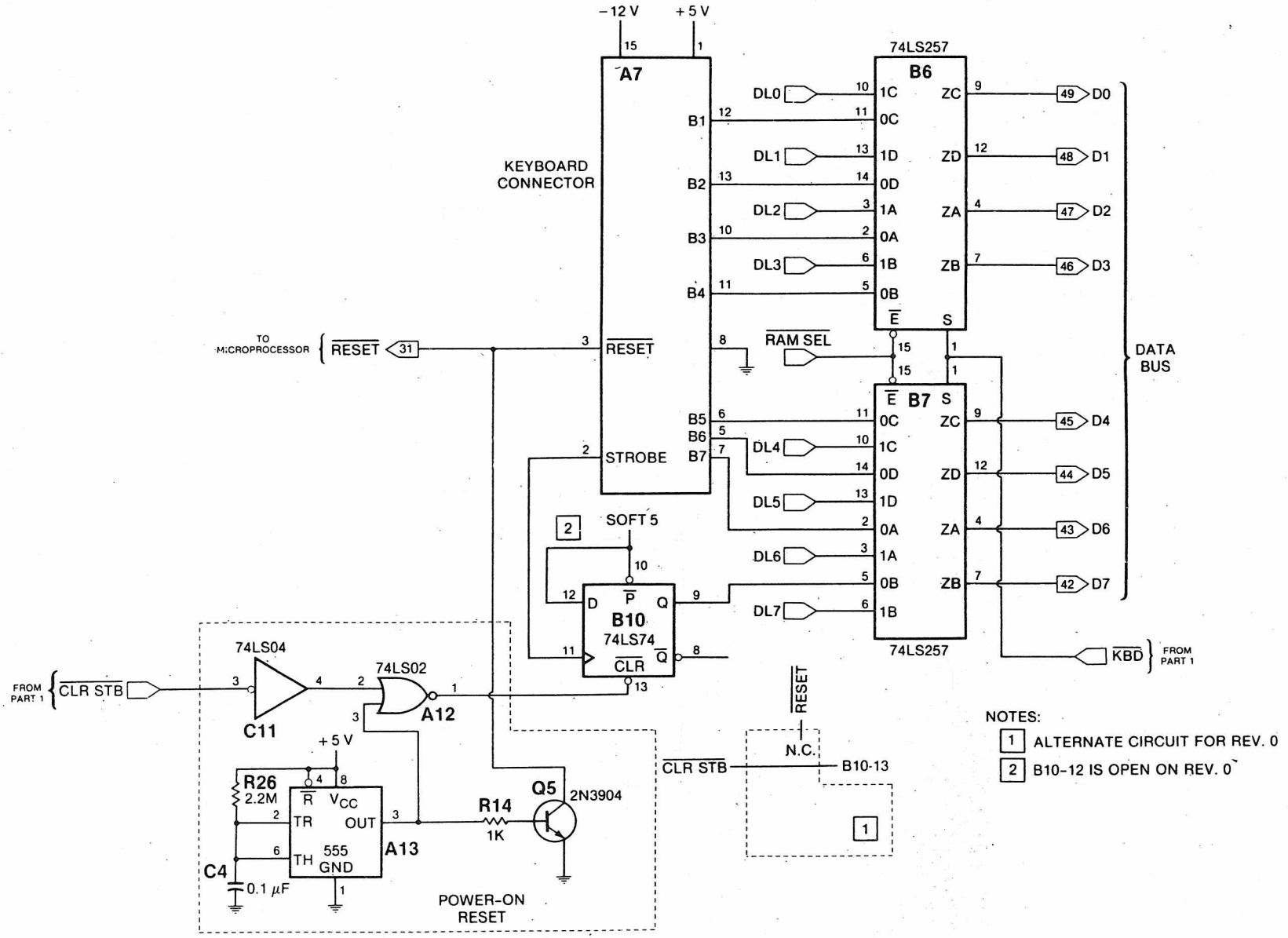



**Fig. C-13.** On-board I/O—part 2 (all revisions).

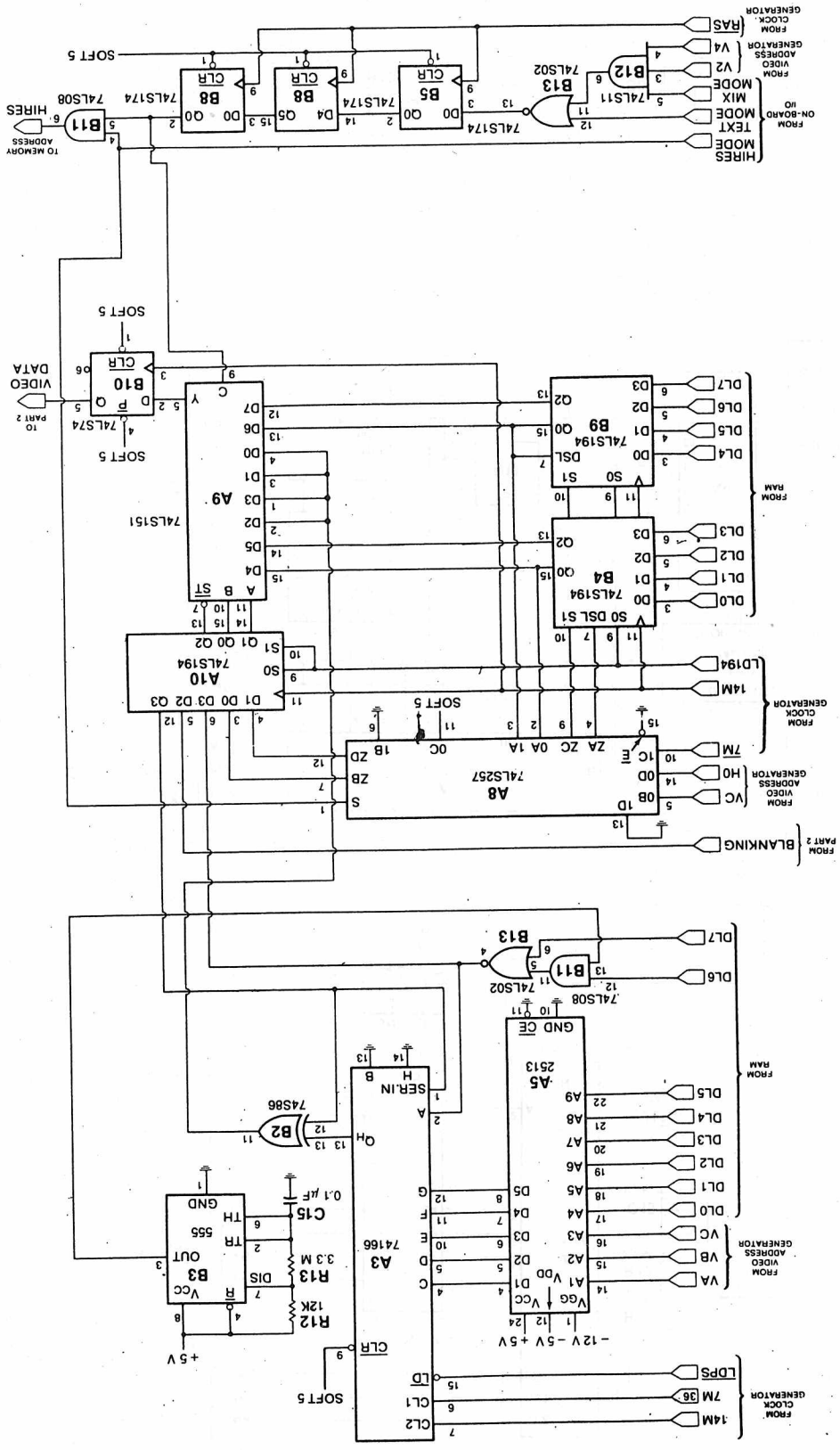



Fig. C-14. Video generator—part 1 (Rev. 0).

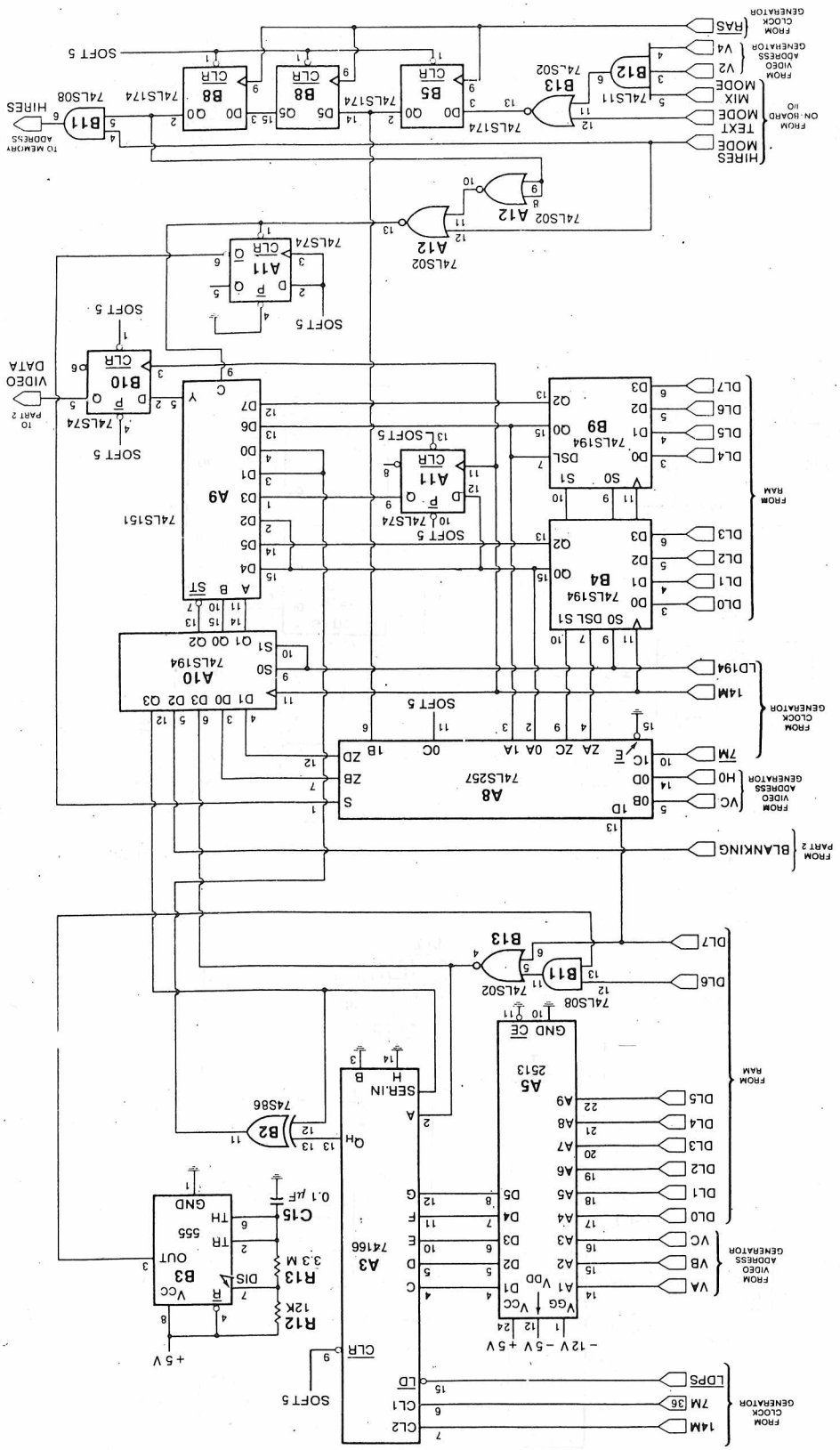



Fig. C-15. Video generator—part 1 (Rev. 1).

**Fig. C-18.** Video generator—part 1 Rev. 7. RPI.

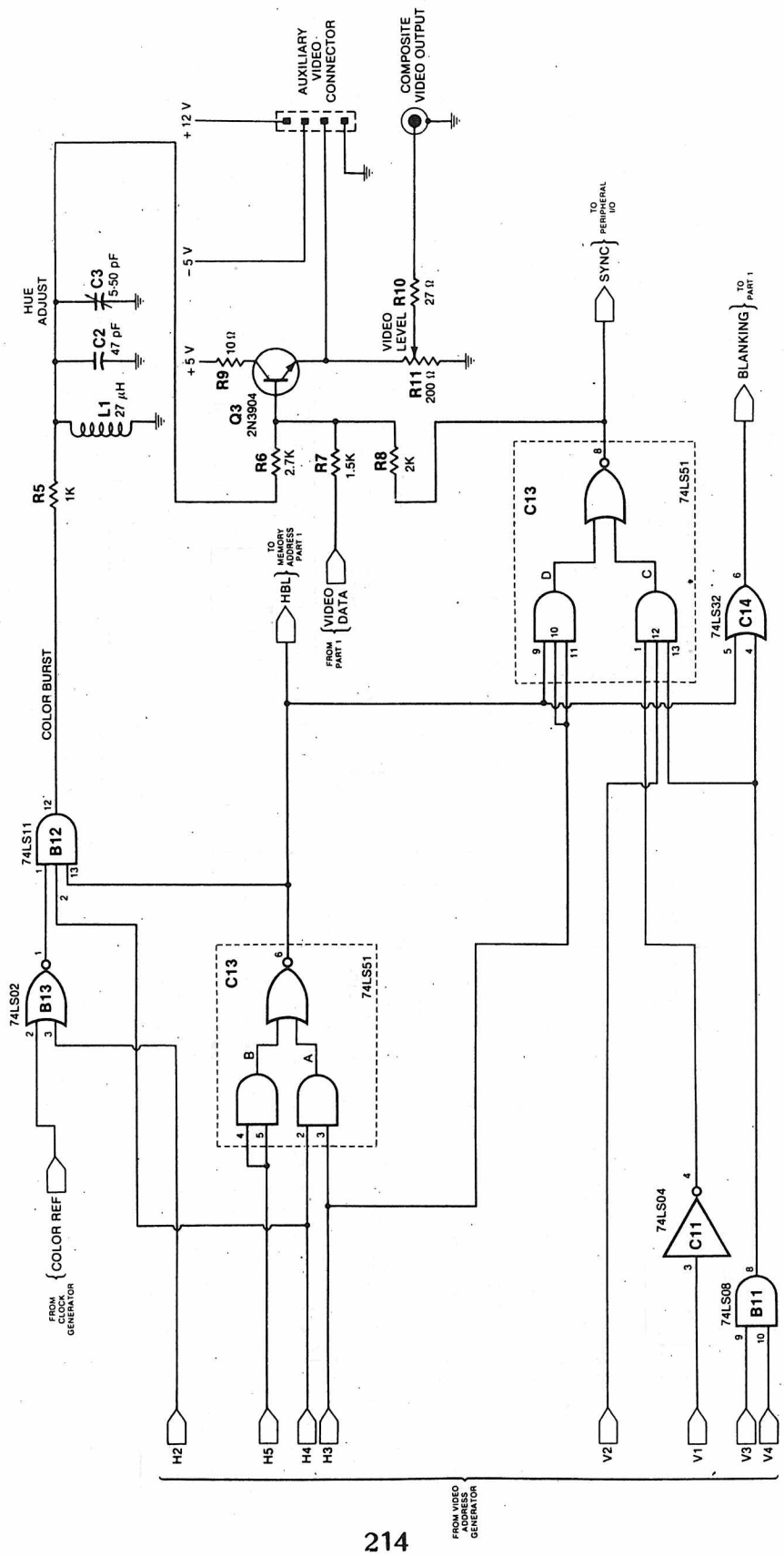



Fig. C-17. Video generator—part 2 (Rev. 0).

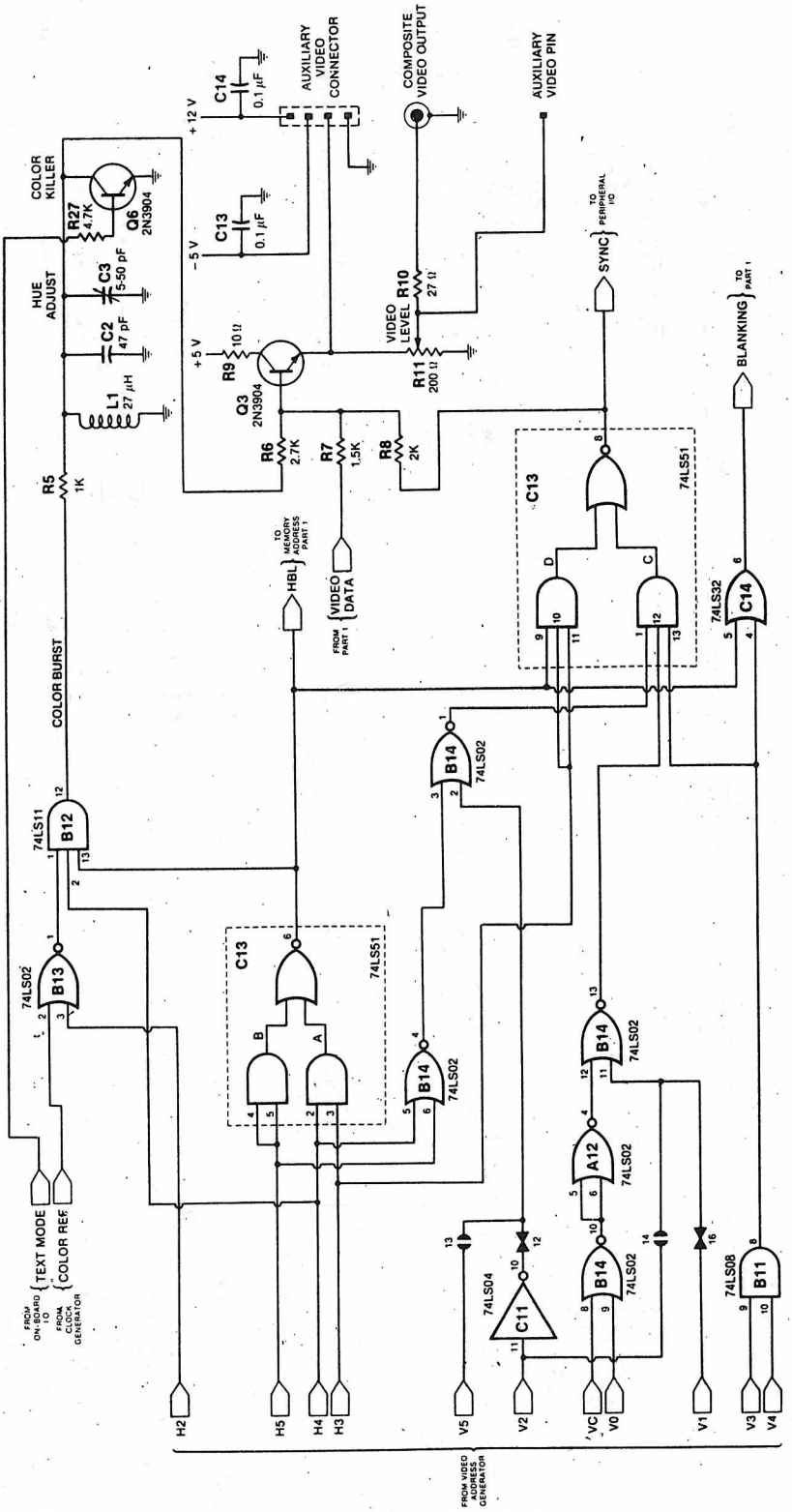



Fig. C-18. Video generator—part 2 (Rev. 1).

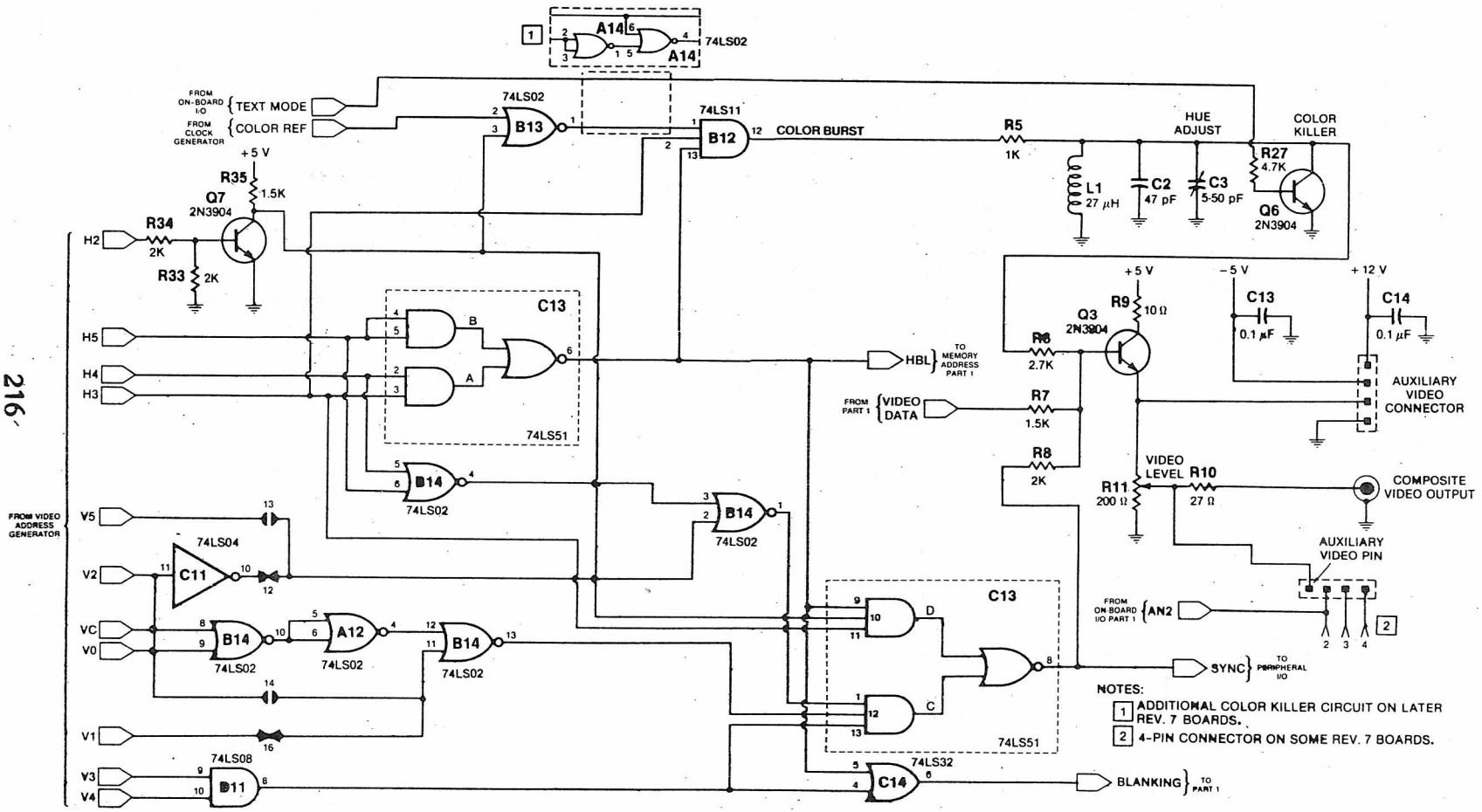



Fig. C-19. Video generator—part 2 (Rev. 7).

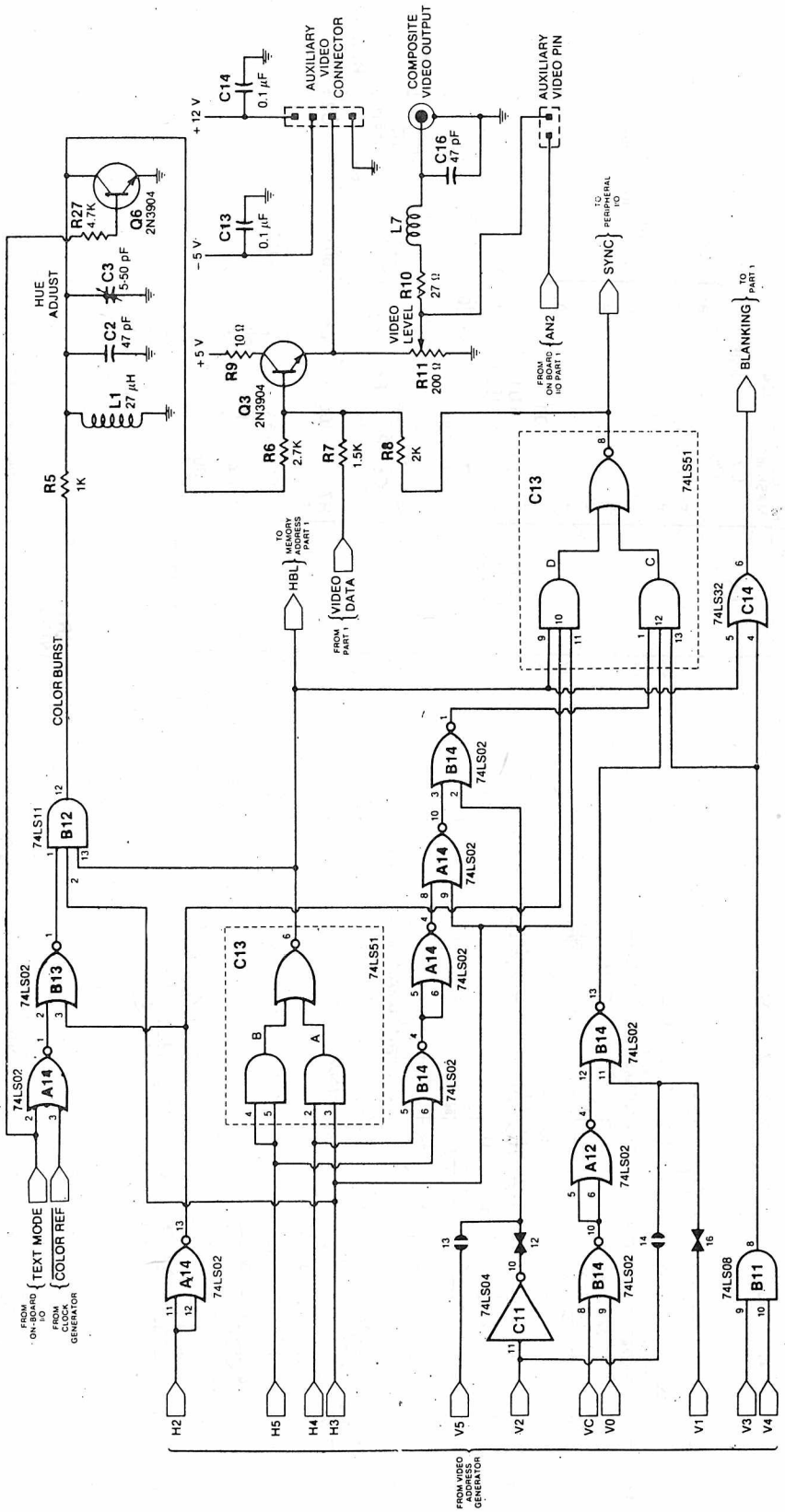



Fig. C-20. Video generator—part 2 (RPI).

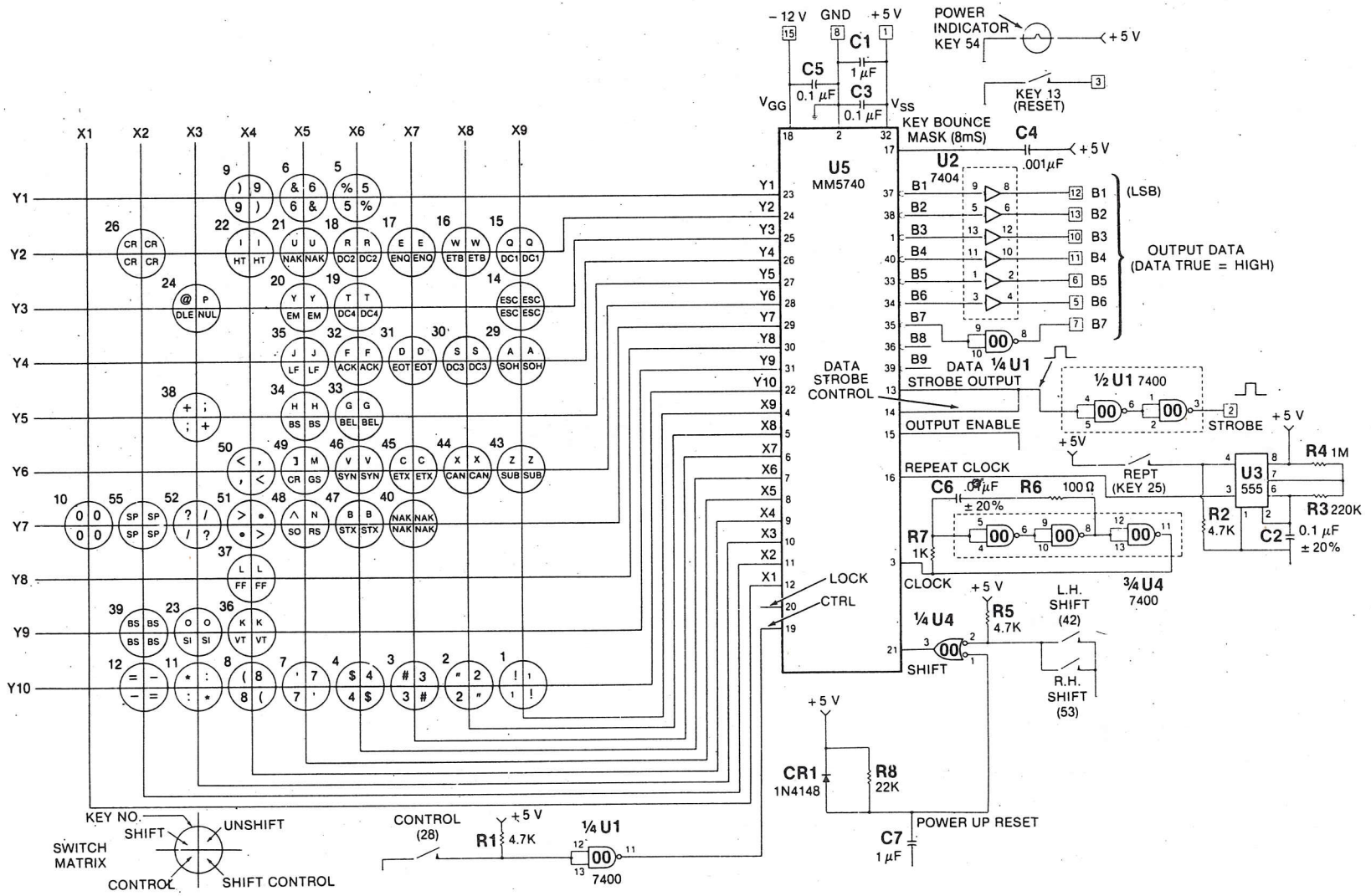



Fig. C-21. Single-piece keyboard.

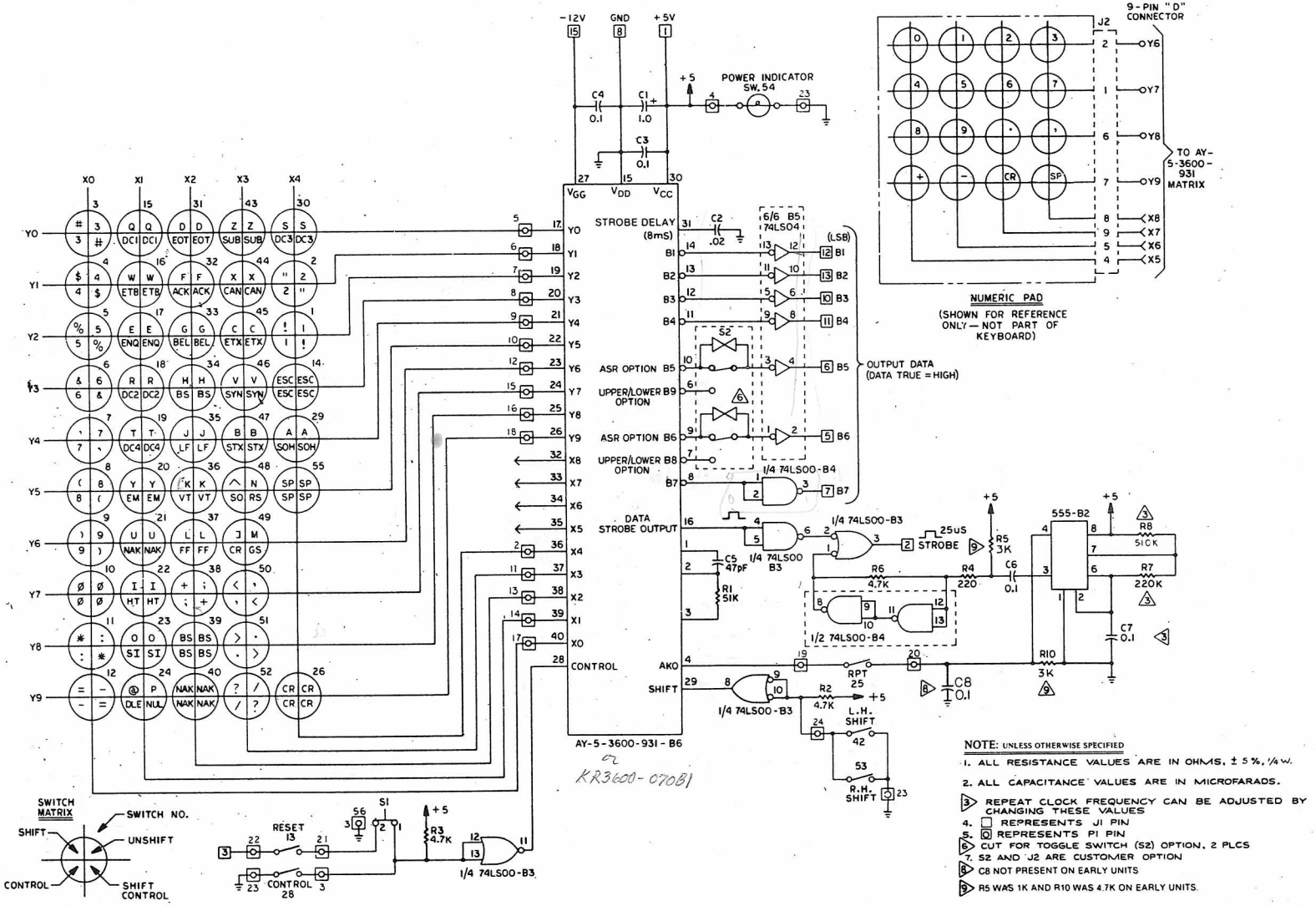



**Fig. C-22.** Two-piece keyboard.

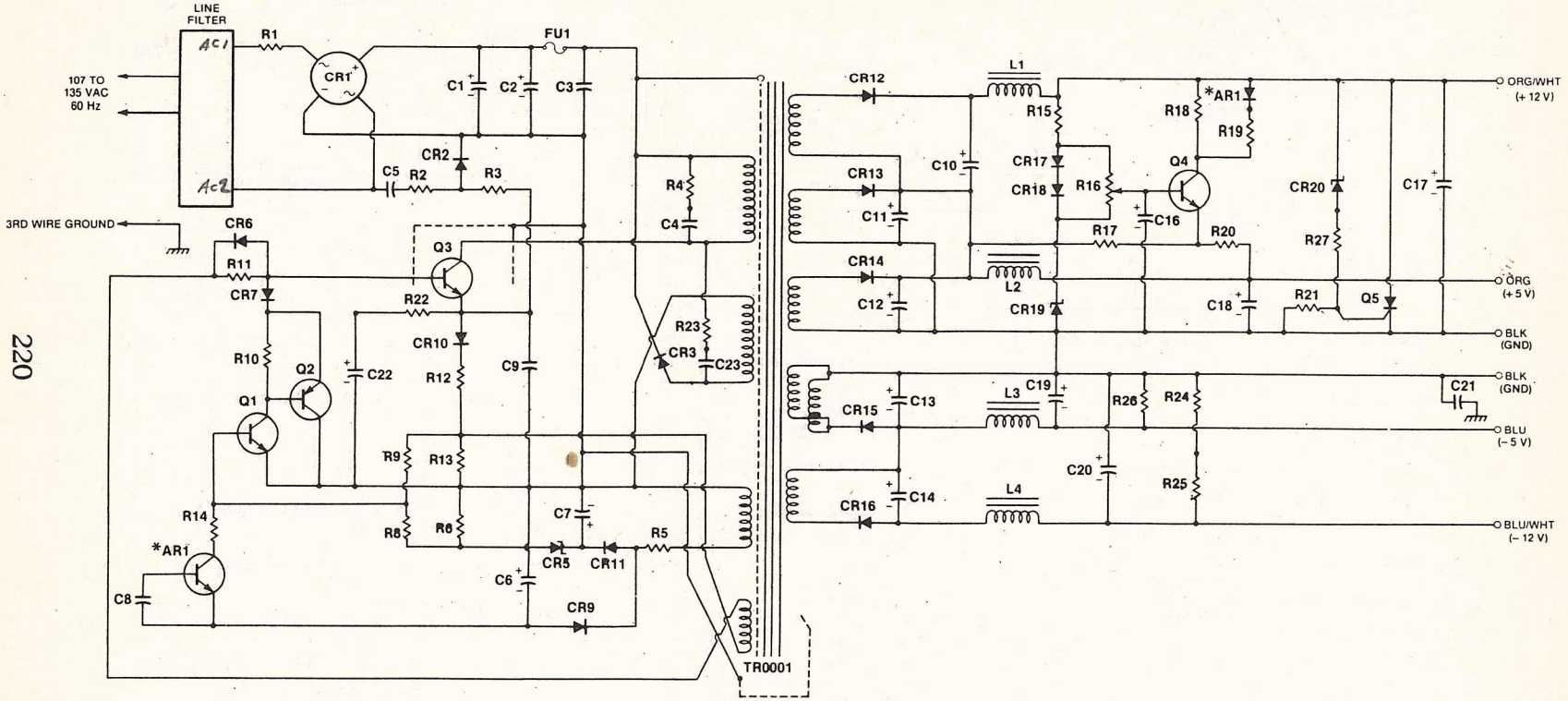



Fig. C-23. Power supply.

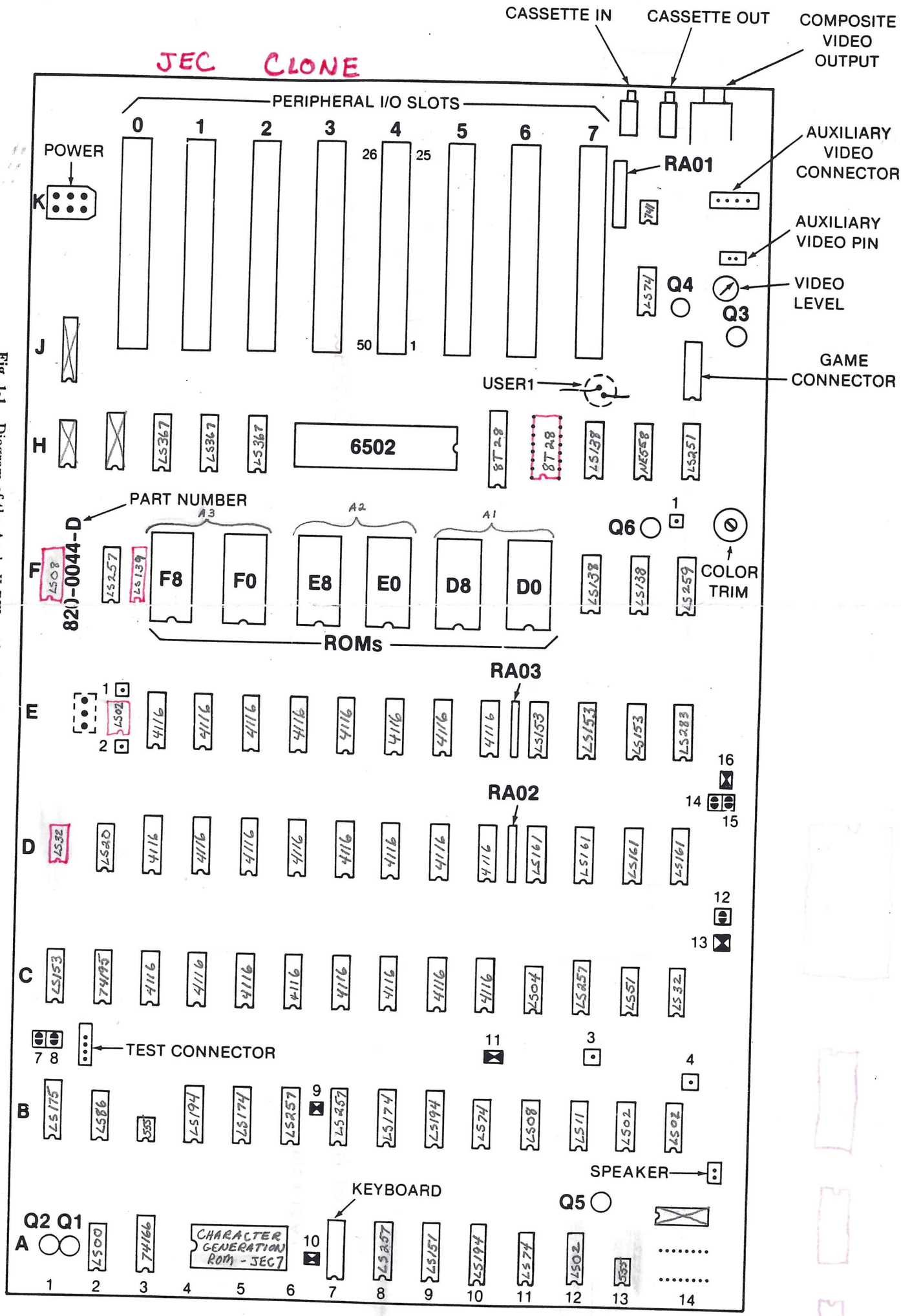
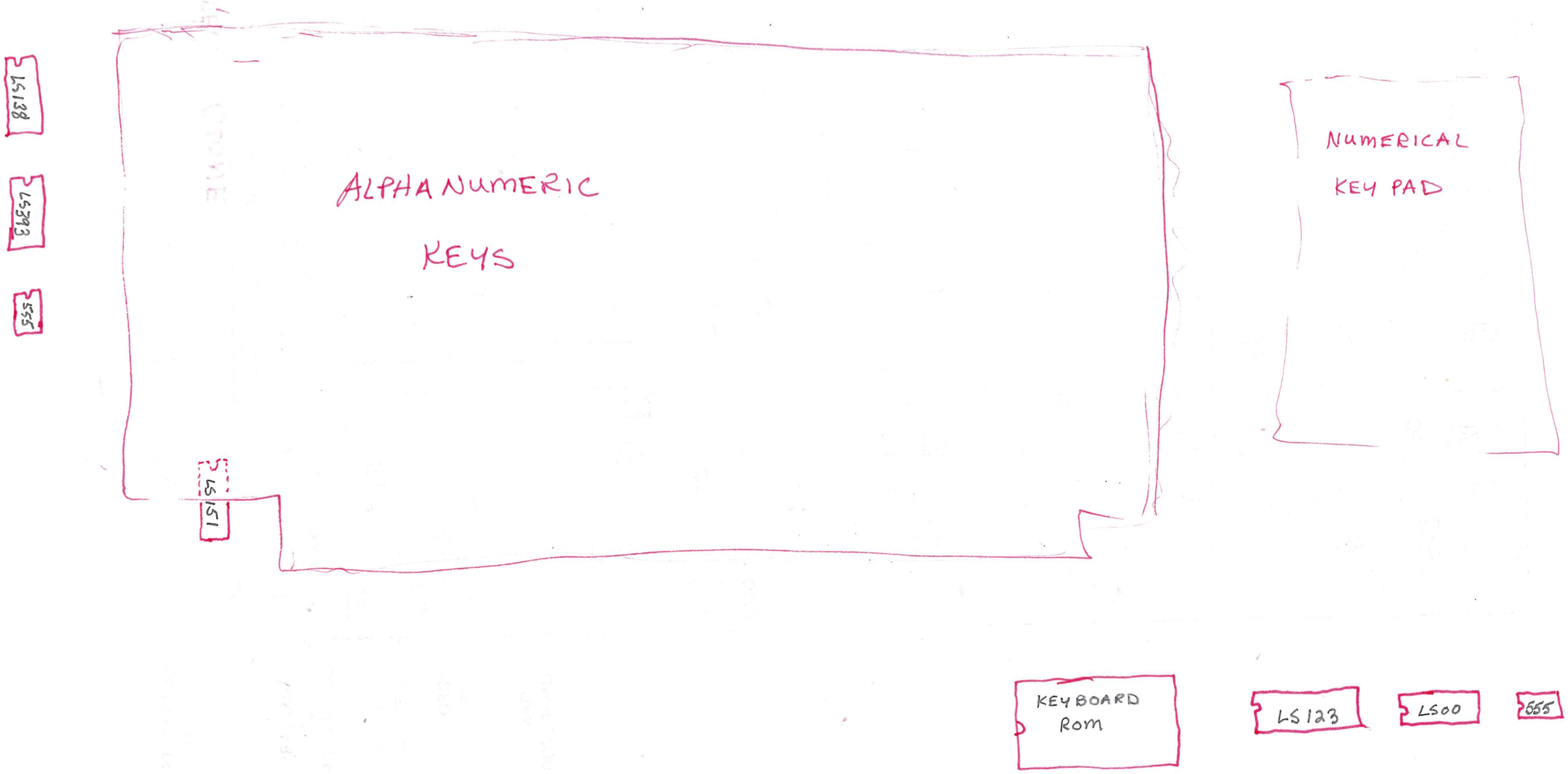



Fig. 1-1. Diagram of the Apple II RFI mother board. (Courtesy Apple Computer Inc.)


LS138
LS393
555
ALPHA NUMERIC
KEYS
LS151
NUMERICAL
KEY PAD
KEYBOARD
Rom
LS123
LS00
555

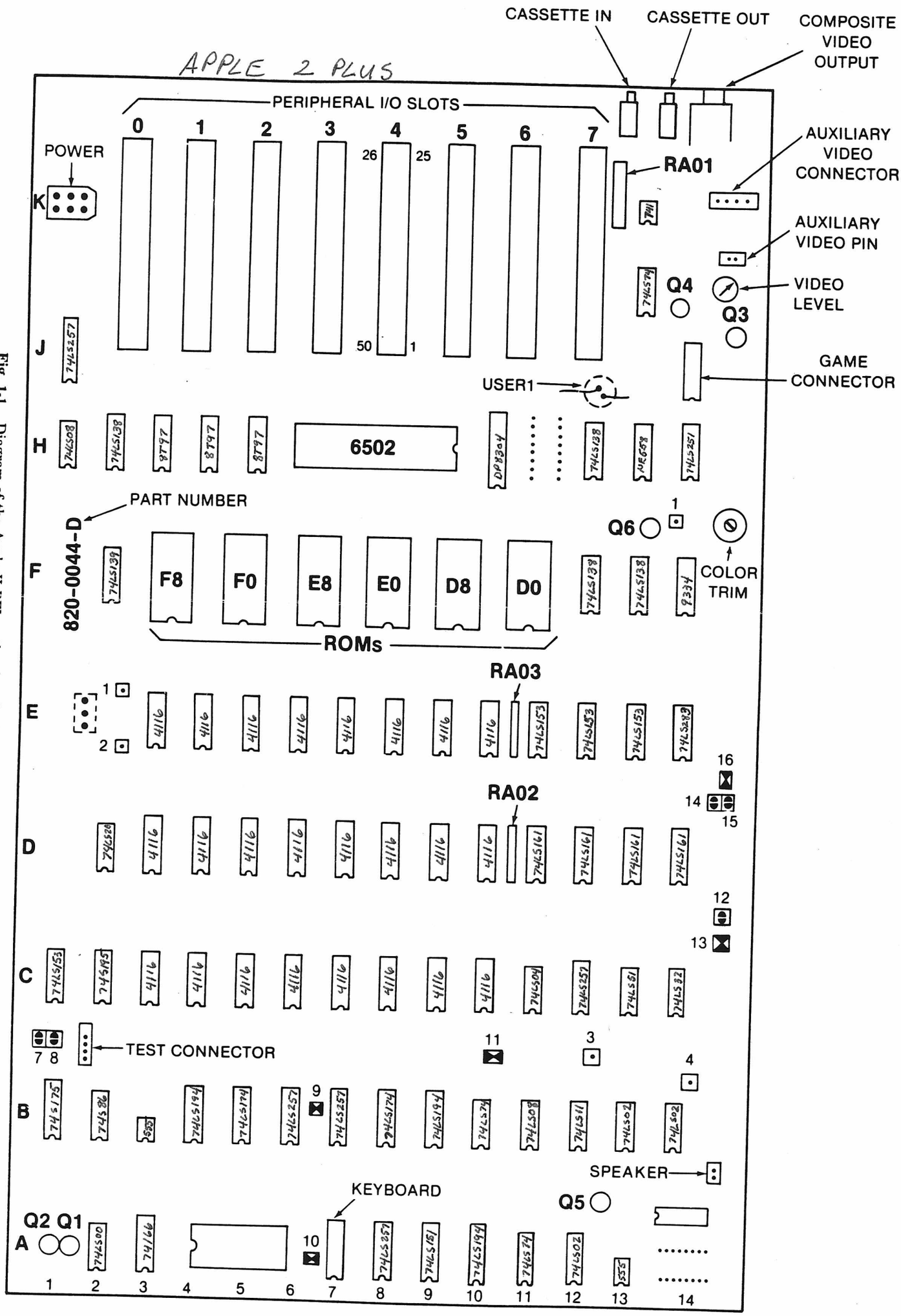



Fig. 1-1. Diagram of the Apple II RFI mother board. (Courtesy Apple Computer Inc.)